

mastertune

The manual for firmware 1.2.1-mastertune v18.3: everything it changes and adds, with the frequency drone in full

mastertune builds on the official Deluge community firmware **1.2.1** and adds a master tune, a lighter audio engine, a CPU monitor, the Deluge as a USB audio interface, cleaner filters and effects, the arpeggiator from 1.3 and a frequency drone with up to 16 tones. This manual explains each change, where to find it on the Deluge and how to use it. Everything not described here works as in 1.2.1.

Written with Claude (AI). mastertune is a personal fork, not a contribution to the official firmware. Please don't report its bugs to the community developers; report them at github.com/Giansn/deluge/issues. Every change is tested in an emulator that runs the firmware's real ARM code, but **v18.x has not been played on a real Deluge yet**. Keep a copy of your SD card.

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1. Getting started

mastertune is a personal fork of the official Deluge community firmware 1.2.1 with a set of changes on top. This chapter tells you which file to install, how to install and check it, and how to go back. Everything this manual does not describe works exactly as in 1.2.1.

1.1 What changes against 1.2.1

Topic	What you get	Chapter
Tuning	A master tune from 415.3 to 466.2 Hz in 0.1 Hz steps, for example 432 Hz. Everything that sounds follows it, external MIDI gear included.	2
Load	About 20–25% less CPU work per sample than 1.2.1, more voices before any are cut, smarter sample streaming, a RAM saver for kits.	3
CPU monitor	One line on the OLED with the load and the voices, plus a warning while the Deluge lowers its quality or cuts voices.	3
USB audio	The Deluge as a stereo audio input on the computer: 24 bit, 44.1 kHz, no driver, bit-exact.	4
Sound	Filters without rustle, steps or clicks, a real shelving EQ, volume in 0.5 dB steps shown in dB, an output limiter, a filter crossing guard, a reworked reverb and delay.	5
Arpeggiator	The arpeggiator of community firmware 1.3, plus latch, ratchet bounce and a ping-pong roll.	6
Workflow	Song browser with versions, countdown on song change, saving without crackle, OLED brightness, flicker-free pad dimming and a list of fixes.	7
Computer	DelugeRec, DelugeTuner, DelugeBaseline, the CPU monitor page, the profiler and file access over USB.	8
Drone	A frequency drone with up to 16 tones, drone tracks, and the drone together with a Korg volca keys.	9, 10

1.2 Which file to install

The current version is **v18.3**. All firmware files are in the folder `mastertune-1.2.1/` of the repository github.com/Giansn/deluge. You need one of these:

File	What it is
<code>deluge-1.2.1-mastertune-v18.3-l2d-3581f019.bin</code>	The current version. With the L2 cache for code and data (§1.5). v18.3 exists only in this variant.
<code>deluge-1.2.1-mastertune-v18-124aeaa2.bin</code>	v18 without the L2 cache, for switching back if the L2 variant crashes or behaves strangely. It lacks the fixes of v18.2 and v18.3.
<code>deluge-1.2.1-mastertune-v17-2cb5e31b.bin</code>	v17 without the L2 cache.

Every earlier version from v2 on is in the same folder (history in §11.7). Next to the newer files lies a `.symbols.json` of the same name. It holds the function names for the profiler (§8.6) and fits only exactly that `.bin`.

1.3 Installing

Keep a copy of your SD card before you install. v18.x has not been played on a real Deluge yet (§1.7).

1. Copy the .bin file into the **root folder** of the SD card. Leave no other .bin file there.
2. Hold **SHIFT** while you switch the Deluge on. The Deluge installs the firmware.
3. Check the version (§1.4).

This is the same procedure as for any Deluge firmware.

1.4 Checking the version

Open **Settings** → **Firmware version**. From v18 on the OLED shows the name in the normal font, on up to three lines and without the hash: 1.2.1-mastertune- / v18.3-l2d. The hash stays in the file name.

Up to v17 the name included the hash, for example 1.2.1-mastertune-v17-l2d-b3385d83.

When you report a problem, always give this name (§11.8).



Settings → Firmware version with v18.3 installed.

1.5 The L2 cache files v17

l2d in a file name means: this firmware uses the processor's L2 cache for code and data. From v17 on this is the main file, and v18.2 and v18.3 exist only this way.

- **Why:** every pass of the audio routine reads more data than fits into the processor's small first-level cache. The L2 cache holds it closer. With v16-l2d the song "New Sitar Grii 10" sounded audibly better on the device, and there was no crash.
- **Not in the numbers:** the CPU comparison with 1.2.1 in chapter 3 comes from an emulator without a cache model. The L2 cache should add to the gain on the device, but that is not measured yet.
- **If it crashes or behaves strangely:** install deluge-1.2.1-mastertune-v18-124aeaa2.bin, v18 without L2, and report what happened (§11.8).

1.6 Going back

To go back, install the other firmware exactly as in §1.3: to v18 without L2 with the file from §1.2, to 1.2.1 with its official file. Your songs keep loading. What happens to the new things:

Saved with mastertune	In an older mastertune version or in 1.2.1
Master tune	1.2.1 doesn't know the entry in CommunityFeatures.XML. When it rewrites the file (after you change a setting, or when you load or save a song with Startup song set to Last opened or Last saved song), a bug in 1.2.1 writes entries it doesn't know back under a wrong name. Back on mastertune the master tune is then at 440.0 Hz: set it again.
Volume in dB	Saved as before. Songs sound equally loud and still load in v17.
Arpeggiator additions	1.2.1 loads the song but skips the kit arp, randomizer, latch and bounce, and loses them at the next save. Walk and Pattern become Up.
Reverb model Digital	The song plays without reverb in 1.2.1.
Reverb damping	Stored as in 1.2.1, so songs stay interchangeable.
Drone and drone tracks	Older firmware skips the drone and the drone rows. Don't save such a song there: the drone rows get lost (chapter 9).
Community settings	v18.3 keeps settings it doesn't know. 1.2.1 and v17 don't: after switching to v17 and back, Output limiter, Filter crossing guard and OLED brightness are back at their defaults.

1.7 How it's tested

- **Reproducible:** every release is built twice, and both builds are byte for byte the same.
- **In an emulator:** the firmware's own machine code for the Deluge's processor (an ARM Cortex-A9) runs in an emulator. It boots, loads songs, renders audio and goes through hundreds of checks and stress tests. Reverb, delay and drone are also measured on the PC.
- **Reviewed:** independent reviewers checked the larger changes with their own test cases.
- **The limit:** the emulator is not the hardware. Timing, the SD card and USB can behave differently on a real Deluge. USB does not run in the emulator at all.
- **On the device:** earlier versions ran on a real Deluge, v16 and its L2 variant among them. **v18.x has not been played on a real Deluge yet.** The listening tests for it are in §5.12.

mastertune is a personal fork, written with Claude (AI), not a contribution to the official firmware. Please don't report its bugs to the community developers. Report them at github.com/Giansn/deluge/issues (§11.8).

2. Master tune

The master tune moves the whole Deluge away from 440 Hz, for example to 432 Hz, or to match an acoustic instrument or an external synth. At 440.0 Hz the firmware behaves exactly like 1.2.1. v2

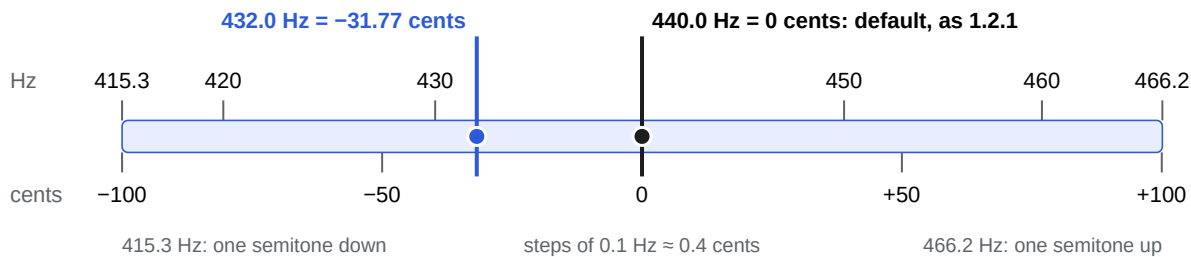
2.1 Setting the master tune

1. Open **Settings** → **Tuning** → **Master tune (Hz)**. The 7-segment display shows TUNE, then MTUN.
2. Turn **[SELECT]** to change the value. The cursor starts on the 1 Hz digit. Move it to the 0.1 Hz digit with the horizontal encoder.
3. Leave the Settings menu with **[BACK]**. The value is saved when you leave it (§2.6).



Settings → Tuning → Master tune (Hz) at 432.0 Hz.

Range	415.3 to 466.2 Hz, that is ± 1 semitone around 440 Hz
Steps	0.1 Hz
Default	440.0 Hz. At this value the firmware behaves exactly like 1.2.1. The only exception: recordings made at another tuning (§2.3).

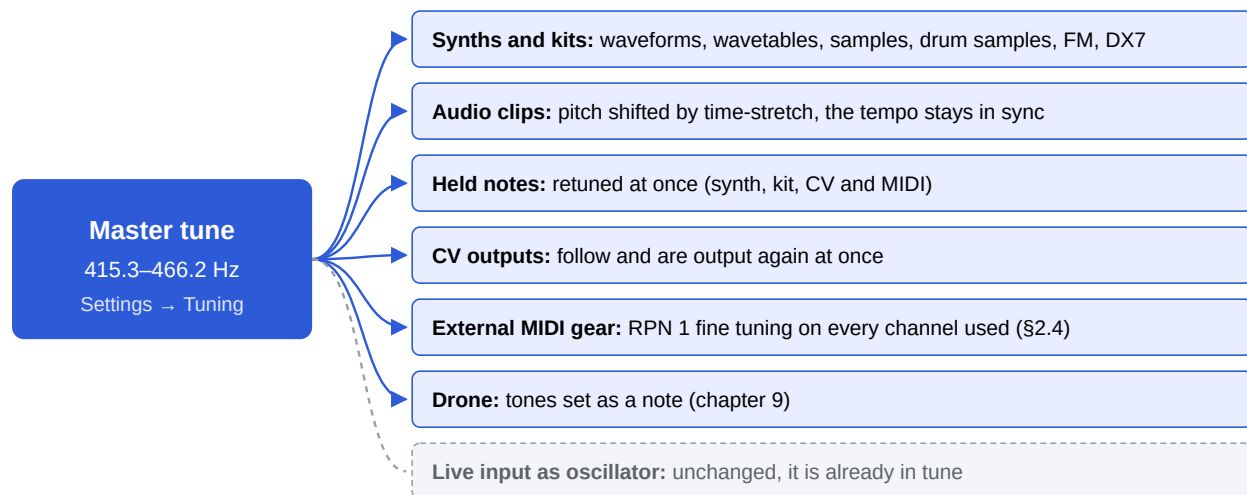


The range of the master tune, in Hz above and in cents below. Pitch is shown on a cents scale, so equal distances are equal musical intervals.

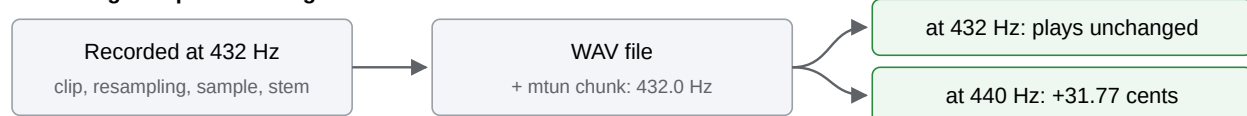
Baroque pitch, A = 415.0 Hz, lies just outside the range. 415.3 Hz, one semitone below 440 Hz, is possible.

2.2 What follows the master tune

Everything that sounds follows the master tune, and held notes change at once when you turn the value. The math is exact: the deviation is below 0.002 cents.



Recordings keep their tuning



Everything that sounds follows the master tune. A recording remembers the tuning it was made in, so it is shifted only by the difference and never tuned twice.

Area	Behaviour
Synth and kit tracks	Waveforms, wavetables, samples, drum samples, FM including its modulators, DX7: all follow.
Audio clips	Follow. The pitch is shifted by time-stretch, so the tempo stays in sync.
Held notes	Retuned immediately when you change the value: synth, kit, CV and MIDI.
CV outputs	Follow and are output again immediately.
External MIDI devices	MIDI tracks and MIDI drums in kits send "Channel Fine Tuning" (RPN 1) on every channel used, with MPE on the member channels (§2.4).
Drone	A tone set as a note follows the master tune (chapter 9). Set the master tune to match an external instrument, for example a volca keys, and the drone fits it (chapter 10).
Live input as oscillator	Unchanged: it already sounds in the current tuning, because the instrument is tuned to it.

2.3 Recordings keep their tuning

- **Tagged when recording:** whatever you record on the Deluge at a tuning other than 440 Hz gets a small mtun chunk (12 bytes) with this tuning in its WAV file. That covers audio clips, resampling, sample recording and stem export.
- **Played back by the difference:** the Deluge shifts a recording only by the difference to the current tuning. A recording made at 432 Hz sounds unchanged at 432 Hz, and at 440 Hz it is raised by exactly +31.77 cents.
- **Root note detection** computes relative to the recording's tuning. Recording, detection, transposition and tuning together hit the target pitch to within 0.001 cents.
- **Other software** skips the chunk and reads the same audio. Tested: Python wave, libsndfile (the basis of many DAWs and editors) and scipy.
- **Files without the chunk**, for example samples from other sources, count as 440 Hz material. Some editors remove unknown chunks when they save a copy. Such a copy then counts as 440 Hz material too.

- **Converted samples:** DelugeTuner writes the chunk too (§8.3). A file whose `mtun` equals the current tuning plays exactly, without resampling.

2.4 External MIDI gear (RPN 1)

The Deluge sends the tuning as RPN 1, "Channel Fine Tuning" (6 control change messages), to MIDI tracks and MIDI drums in kits:

When	What is sent
You change the value	Immediately, to all channels the song uses. This also retunes held notes.
After play start	Once more, for devices switched on in the meantime. It goes out after the first notes, so it never delays them.
Before the first note on a channel	When the channel doesn't have the current tuning yet, for example after loading a song with new channels.
Shared channels	If several tracks or drums share a channel, it is tuned only once.
Never detuned	If no tuning other than 440 Hz has been active since power-on, nothing extra goes out. After a return to 440 Hz the devices are reset to 0 cents once.

The DIN MIDI output has no overflow protection. The Deluge therefore sends only while there is room and the rest a few milliseconds later. Even fast turning with many MPE channels produces no MIDI garbage.

- **Your device must evaluate RPN 1.** Many synths and DAWs do, but not all.
- RPN 1 covers -100 to $+99.99$ cents. Only at the two extremes, 415.3 Hz (-100.02 cents) and 466.2 Hz ($+100.13$ cents), a deviation of at most 0.15 cents remains.
- A device switched on later gets the tuning a few milliseconds after the next play start. If it applies the tuning only at note-on, the very first notes of that run still play at 440 Hz.

2.5 CPU cost and sample libraries

Away from 440 Hz the Deluge resamples every sample voice, even an untransposed drum hit, just like a transposition by a fraction of a semitone.

- **Cost:** with sinc interpolation a resampled stereo voice costs about six times as much as direct playback. Audio clips run through the time-stretcher all the time. That is the biggest item, noticeable only with many audio clips at once. Recordings made at the current tuning play natively.
- **Cheaper since v3:** every retuned, transposed or time-stretched sample voice without a cache hit costs about 40% less than in 1.2.1.
- **Often small in practice:** in the emulator "New Sitar Grit 10" costs only 0.2% more at 432 Hz than at 440 Hz, because its samples play transposed anyway.
- **Sound quality:** the pitch shift uses the same interpolation and time-stretcher as transposing. Samples in "Pitch/speed independent" mode and audio clips therefore have the usual, very quiet time-stretch artifacts.

To remove the cost, convert your sample library once to the master tune with DelugeTuner (§8.3). The samples then play directly again. On the test card in the emulator the whole card needed 34% less work without the sample cache, 6% less with it.

2.6 Where the setting is stored

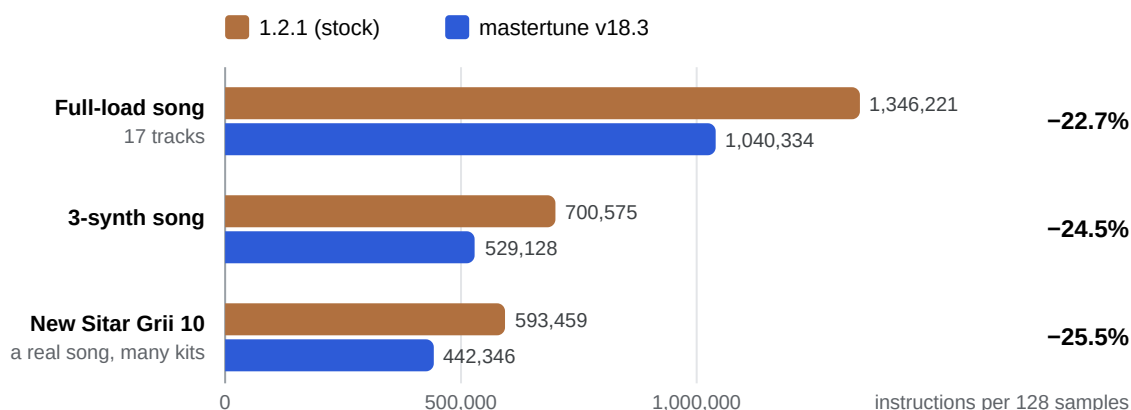
- In `CommunityFeatures.XML` in the root folder of the SD card, entry `masterTune`, in tenths of a Hz: `4320` is 432.0 Hz. It is saved when you leave the Settings menu.
- 1.2.1 doesn't know the entry, and a bug there loses it when it rewrites the file (§1.6).

3. Lighter load and the CPU monitor

mastertune does the same work with fewer instructions and wastes less time between audio blocks. The CPU monitor shows you the load, the voices, and when the Deluge has to save on quality or cut voices.

3.1 Measured against 1.2.1

Stock 1.2.1 and v18.3 rendered the same songs in the emulator, both at 440 Hz. The first measure counts the instructions the Deluge needs per window of 128 samples (2.9 ms). That is its DSP work.



DSP work per 128 samples in the emulator, stock 1.2.1 against v18.3, same songs, same settings. Shorter is lighter.

The second measure runs the firmware's own task manager and records how much of the time goes into the audio routine:

Song	Measure	1.2.1	v18.3	Change
Full-load song	instructions per 128 samples	1,346,221	1,040,334	-22.7%
same, with culling as on the device	voices on average (voices cut in 4 bars)	20.4 (380)	32.0 (113)	+57% voices
3-synth song	instructions per 128 samples	700,575	529,128	-24.5%
"New Sitar Grii 10"	instructions per 128 samples	593,459	442,346	-25.5%
same, IPC 1.0	share of the time in the audio routine	94.7%	43.8%	-54%
same, IPC 0.8	same	97.1%	54.8%	-44%
same, IPC 0.7	same	97.3%	97.5%	±0
3-synth song, IPC 1.0	same	97.1%	92.1–94.8%	-2 to -5%

IPC is the number of instructions per clock cycle. The value on the device is unknown. Lower values stand in, roughly, for cache misses and memory waits. At IPC 0.7 the Sitar song lowers its quality in 32% of the renders with v18.3 and in all of them with 1.2.1.

What it means:

- **About 20–25% less DSP work per sample**, the same in all three songs.
- **At moderate load the audio routine's time can roughly halve** (the Sitar song: -44 to -54%). 1.2.1 computed in tiny blocks with a fixed overhead per track (§3.2). Near full load this advantage disappears: both then use nearly all the time and differ only in how often they lower the quality and cut voices.
- **More voices before cutting**: 32 instead of 20.4 on average in the full-load song. Part of it comes from the lighter DSP, part from the gentler culling taken over from the current community firmware.
- **"Half the load" is not right as a general claim**. It holds only at moderate load.

Caveats. The emulator has no memory or cache model, so the L2 cache is not in these numbers; on the device it should add to the gain. The audio of v18.3 is not bit-identical to 1.2.1, because v18 changed filters and saturation (chapter 5); voices and level match. Streaming songs, 432 Hz and UI or MIDI load were not part of this comparison.

3.2 Where the savings come from

Change	Since	What it does
Scheduling fixed	v17	A calculation error in 1.2.1 set the playback task's interval to 0. It was always due, so the task manager called the audio routine again about 10 μ s after each pass, each time for only 4–12 samples, and every pass went through all tracks. Now, up to 65% load, it computes in blocks of at least 60 samples. Lowered quality and card accesses switch this minimum window off.
Silent tracks skipped	v4, v17	A kit or audio track that has output only silence for 4096 samples skips its effects chain, as long as mod FX, delay, stutter, sample rate reduction, saturation and compressor are off and nothing records. v17 checks this before setting up the chain. The sound is bit for bit the same.
Gentler culling	v4	As in the current community firmware: less often, fewer voices, and half of them faded out instead of cut.
No needless cuts while streaming	v17	Culling and quality reduction judge the audio routine's own CPU time, no longer the loading task including the wait for the card.
Faster DSP	v3, v13	Resampled sample voices about 40% cheaper; faster filters, oscillators and track effects. All bit-identical.
L2 cache	v17	For code and data, in the main file (§1.5).

The price of the minimum window. Under light load, notes you play live start with the next block, on average about 0.5 ms later. The reserve against dropouts is smaller: at least 63 samples (1.4 ms) in the emulator, with no dropout in any test.

3.3 Sample streaming and the Kit RAM saver v9

The SD card does not get faster, but its throughput now goes where it is needed:

- **Loading by urgency.** 1.2.1 sorted its load jobs by memory address instead of priority. Now what a playing voice needs next comes first. You get fewer voices breaking off and fewer "card too slow" messages under load, for example when you change kits while playing.
- **Double reserve.** A playing voice keeps three clusters ahead instead of two: with 32 KB clusters about 250–370 ms instead of 120–190 ms (stereo). It costs 32 KB of RAM per voice that is streaming.
- **Short samples stay in RAM.** Samples of up to four clusters (128 KB with 32 KB clusters) now really stay in memory, as intended in 1.2.1.
- **No lost hits.** If a sample's start is not in RAM when you strike it, the voice waits until it is loaded (usually a few milliseconds, at most 100 ms) and plays from the beginning. 1.2.1 dropped the hit.

Kit RAM saver: Settings → Community features → **Kit RAM saver** (7-segment KRAM), on by default. Kit rows without notes in all of the song's clips release the held start of their samples, usually 64 KB per sample. A kit with 50 samples of which the song uses 5 frees around 3 MB. The samples stay in the kit. When a row gets notes, the Deluge fetches its start back within one second. Not affected: kits with the kit arpeggiator or with MIDI learn for the whole kit, and rows with their own MIDI learn, because they can play without notes.

If you audition a row that has no notes, or play it over MIDI, its very first hit can come a few milliseconds late. Sequenced notes are not affected. If this bothers you, switch the Kit RAM saver off.

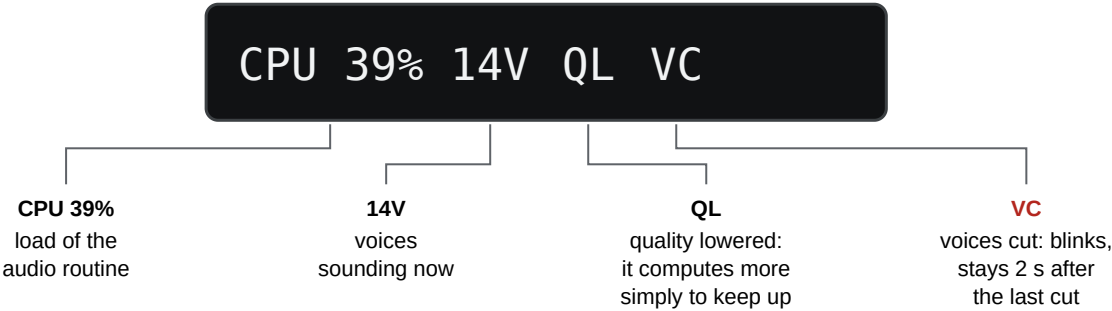
3.4 The CPU monitor

The CPU monitor puts one short line at the top left of the OLED, over whatever view you are in.

- **Menu:** Settings → **CPU monitor** (7-segment CPU): Off, On, Alerts or Profile.
- **Shortcut:** hold **LEARN** and press the **TEMPO** knob.
v17



The monitor's line over song view: 39% load, 14 voices.



QL and VC appear only while it happens. The mode Alerts shows only them.

The four parts of the monitor's line. The example shows all of them at once.

Mode	What you see
Off	Nothing. Switched off, the monitor costs a single check per pass of the audio routine.
On	The line, for example CPU 39% 14V, plus QL and VC while they happen.
Alerts	Only QL and VC, and only while they happen.
Profile	The line as with On. In addition the Deluge sends measurements for the profiler to the computer (\$8.6). v16

- **The shortcut** switches between Off and the mode you used last (On or Alerts), in every view and also while playing. It confirms briefly, on the 7-segment display C-ON, C-OFF or C-AL.
- **Saved at once:** the mode goes to the card immediately (CommunityFeatures.XML) and applies again after a restart. Profile is chosen in the menu; after a restart it counts as On. The startup song is left untouched.
- **The real load:** from v17 on, passes that compute nothing don't count as busy. Up to v16 an idle Deluge showed 85–90%. That was busy waiting, not overload.
- **To the computer:** with the monitor on, the Deluge sends one message per second on USB MIDI port 3 only. DIN MIDI never gets anything. If the USB send buffer is full, the message is skipped, so notes and clock keep priority. Chapter 8 shows how to log it.

3.5 When QL or VC appear

QL means the audio routine is behind the output. The Deluge then saves on quality: simpler oscillator tables, linear instead of sinc interpolation. From about 80 samples behind it switches voices off: that is VC. What costs CPU:

What	Cost and what you can do
Master tune away from 440 Hz	Every sample voice is resampled. Convert the library once with DelugeTuner (§8.3).
Drone	A binaural tone 0.6%, with Life, FM and Pulse together 2.1%; 16 such tones a good 30%. The drone is not culled like synth voices, so use many living tones with an eye on the monitor (chapter 9).
Drone tracks	0.3–0.45% per sounding row. Rows with a closed gate cost nothing.
Song reverb	Computed once for the whole song. Digital needs about a third more than Mutable.
Delay	Since v14 about 0.8% more per delay at rest and 1.1% more during a time change. A modulated time costs more, because the delay then resamples.
Drive LPF	Oversampled only high up with resonance (§5.4). Under load the oversampling switches off.
EQ	Since v18 twice as much as before with both bands in use.

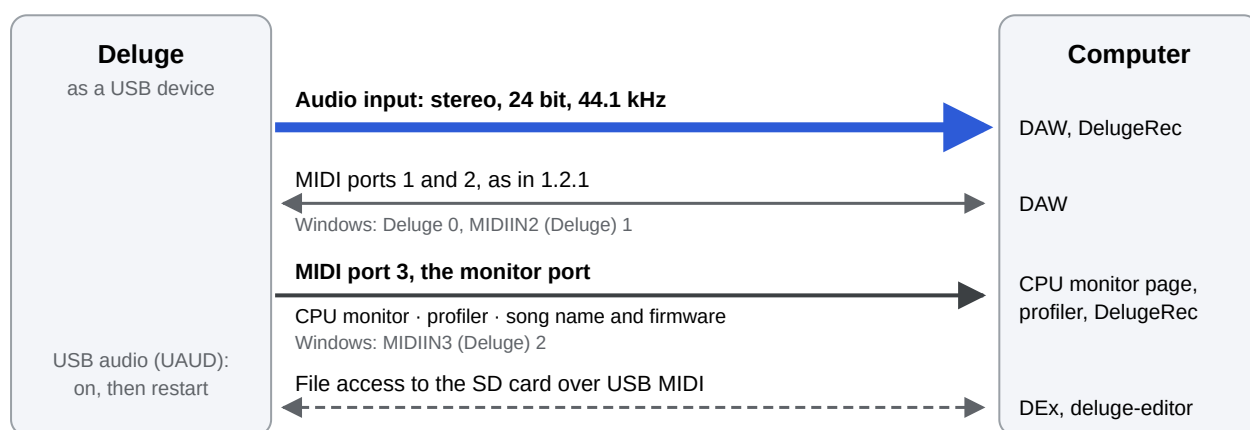
The percentages are estimates from the emulator at 1 instruction per clock cycle. They are good for comparisons, not exact for the device.

4. The Deluge as a USB audio interface

Connect the Deluge to a computer by USB, and the computer sees a stereo audio input next to the usual MIDI ports. It receives exactly what is at the Deluge's outputs, bit-exact, without a driver. v8

4.1 What the computer receives

- **Stereo, 24 bit, 44.1 kHz**, as a standard USB audio device (USB Audio Class 1.0). No driver needed.
- **The Deluge's output signal, bit-exact**: the same signal the Deluge records when resampling, with the master volume and input monitoring. The drone is in it too (chapter 9).
- **The Deluge controls the level**. The computer has no volume control for this input. The Deluge's analog VOLUME knob acts after the converter, so it doesn't change the USB signal either.
- **One direction only**: from the Deluge to the computer. Audio from the computer to the Deluge is not there yet.



What runs over the one USB cable. Audio goes only from the Deluge to the computer. Port 3 carries the CPU monitor, the profiler and the song name for DelugeRec.

4.2 Switching it on

1. Open Settings → Community features → **USB audio** (7-segment UAUD) and set it to on.
 2. Restart the Deluge. The setting only takes effect at startup, and only when the Deluge is connected to the computer as a USB device, not as a USB host.
- From v16 on the Deluge saves this setting immediately. Up to v15 it was saved only when you left the menu: leave it with BACK before you restart.
 - Off is the default. Then the Deluge behaves as without USB audio.

4.3 On the computer

System	Where the Deluge shows up
macOS	Audio MIDI Setup shows "Deluge" with 2 inputs.
Windows	Settings → System → Sound → Input: "Deluge".
Linux	<code>arecord -l</code> lists "Deluge".
DAW	Choose "Deluge" as the input and set the project to 44.1 kHz.

To record without a DAW, use DelugeRec (§8.2). It names each take after the song.

4.4 Timing and latency

- **The Deluge sets the clock.** Each USB packet carries 44 or 45 frames, depending on how full its buffer is. If its crystal drifts against the computer's clock, it sends one frame more or less. Nothing is resampled; the samples arrive bit-exact.
- **Latency:** about 6 ms of buffer plus 1–2 ms for USB.
- **Hiccups:** if the computer stops reading for a while, the Deluge discards the stale buffer and starts again after 6 ms of silence. If the audio engine stalls, a short silence comes instead of crackles.
- **MIDI is not affected** by the audio stream.

4.5 The song name for recordings v18

While the computer has the audio stream open, the Deluge reports the song name and the firmware version on USB MIDI port 3. DelugeRec uses this to name its takes, for example "Song name, 28.09.2026 - 1.2.1 v18.3.WAV" (§8.2). The Deluge only sends this; it never waits for an answer. Because USB doesn't run in the emulator, this has to be confirmed on the device.

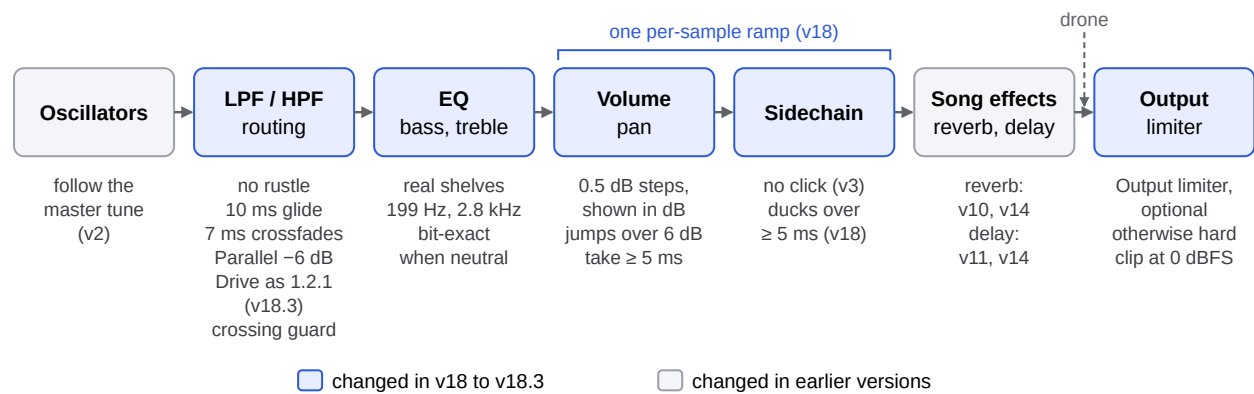
4.6 Limitations and messages

Situation	What to know
Other sample rates	Only 44.1 kHz. If the DAW runs at another rate, the operating system resamples. In exclusive mode or with ASIO the project must be at 44.1 kHz.
MIDI ports moved	With USB audio on, the computer sees the Deluge as a new device. You may have to assign the MIDI ports in your DAW again.
Not recognised or hangs	Unplug the USB cable, start the Deluge, switch USB audio off.
UGAP, "USB audio gap"	The computer got an empty packet: a 1 ms gap in the recording. This can happen under very high load, because the Deluge disables all interrupts while it computes each track. The message comes at most every 10 seconds. Please report it (§11.8).

5. Sound: filters, EQ, volume and effects

v18 reworks how a sound moves through the filters, the EQ and the volume stages. Every change now glides sample by sample, nothing steps or clicks, and the level controls speak dB. v18.2 and v18.3 fix what the first v18 missed. The reverb and the delay were reworked earlier, in v10, v11 and v14.

5.1 The signal path



The path of a sound, simplified, and what changed on the way. Volume, pan and sidechain share one ramp that moves per sample. The drone joins after the song effects.

5.2 Filters without rustle, steps and clicks v18

Change	What you hear
No rustle at low cutoff	The ladder filters (12 dB, 24 dB, Drive) carried a deliberate "analog noise" on their cutoff, already in the original firmware: 0.44% of fluctuation, audible as a band of noise around every note. It is gone.
Silence after the music	Below 4.5 kHz the ladder computes with more precision. When the music stops, the filter really goes silent.
Resonance at low cutoff as in the original	Below about 310 Hz, the 12 and 24 dB ladders lower their resonance again by up to 9%, and song, kit and track filters saturate as in the original. Self-oscillation there is 6 dB quieter, and the hum at 32 Hz (for example in "New Sitar Grii 10") disappears.
Per sample, not per block	Cutoff, resonance, morph and the level compensation move sample by sample. Large jumps pass through real intermediate settings. The zipper noise drops by 7 to 72 dB, depending on the filter. A filter that stands still computes bit for bit as in v17.
10 ms glide	Every new filter value glides over 10 ms, wherever it comes from: knob, automation, MIDI or patch cable, in voices and in song, kit and track filters. Before, only the knob in synths glided.
Crossfades	7 ms when you change the mode or the routing and when you switch a filter off; a linear fade-in when you switch it on. The clicks drop from up to 254 times to 1.7 times the normal step between two samples.
Fresh voice state	A new note inherits nothing from the previous one.
After a silence v18.2	A kit or audio track that comes back in after a pause starts right away with its current filter, EQ and volume. v18 still ran old movements over the first milliseconds: a short tick or a brighter attack. If you installed v18, take v18.3.

Parallel routing is 6 dB quieter. With routing **Parallel**, a filter that is switched off no longer adds dry signal, and the sum of both filters has half the volume (−6 dB). Songs with this routing therefore play 6 dB quieter than before v18. Raise their volume if you miss the level.

5.3 Resonance and self-oscillation

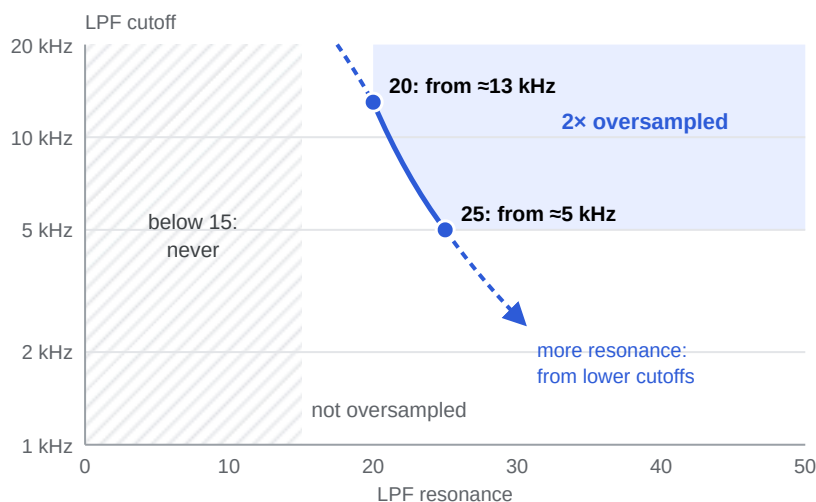
The filters can still sing by themselves at high resonance, as in 1.2.1 and in the original firmware. The firmware does not limit the resonance.

- **HPF whistle** [v17](#): with a lot of resonance the HP ladder self-oscillates. Since the community's filter rework it limited 12–18 dB higher than the original firmware; in the song master that gave a whistle up to 27 dB above the music. It now limits like the original again: about 17 dB quieter in song, kit and track, about 12 dB in synths. The tone is not completely gone; it sits at about the level of the music. Below about 39% HPF resonance the sound is bit for bit the same as before.
- **24 dB ladder**: it self-oscillates flat, as in 1.2.1: −19 cents at 600 Hz, −52 at 1.9 kHz, −127 from 5.7 kHz. The 12 dB ladder hits the pitch.
- **Drive LPF**: it self-oscillates from about 50% resonance, at around 9–21 kHz, as in 1.2.1. Below the oversampling threshold its tone is 56 cents flat at 2 kHz (§5.4).

Example: "the LPF has no effect". In one song the song LPF's resonance was at maximum. From about 36 on the cutoff knob, the filter oscillated by itself, about 10 dB above the music, and its tone followed the knob down. Fix: in song view, with the filter button on LPF, turn the lower gold knob (resonance) below about 35, then save. To delete an automation of the filter as well, hold **[SHIFT]** and press the upper gold knob ("Automation deleted"); without **[SHIFT]** the press changes the filter type.

5.4 Drive [v18, v18.3](#)

- **The bass stays**: with resonance, Drive now loses at most 0.53 dB of bass. Before v18 it lost up to 13 dB.
- **Real 2× oversampling** with half-band filters: the mirror images drop to −88 dB. 1.2.1 only doubled samples.
- **Only where it matters** [v18.3](#): Drive oversamples where 1.2.1 did, high up with resonance. There, resonance and saturation would otherwise alias audibly.
- **Switching is crossfaded**. A little hysteresis keeps an LFO that moves around the threshold from switching back and forth.
- **Under load** the oversampling switches off, as in 1.2.1.



Above the line

2× with real half-band filters, images at −88 dB switching crossfaded, with a little hysteresis off under load, as in 1.2.1

Below the line

as in 1.2.1: self-oscillation 56 cents flat at 2 kHz

Where Drive runs 2× oversampled in v18.3: the same place as in 1.2.1. The two points and the limit at resonance 15 are from the tests; the lines between and beyond them only guide the eye.

Why v18.3 changed it. v18 oversampled Drive at every setting. A song with many Drive filters then needed 3.5 times real time at the start of the bar (1.2.1: 1.7 times) and lost almost all its voices before the load protection could step in. In the stress test with every LPF on Drive, v18 kept 6 voices on average, v18.3 keeps 15 and v17 kept 20. The price: below the threshold Drive self-oscillates as in 1.2.1 again, 56 cents flat at 2 kHz (v18: 30 cents).

Drive's half-band filter delays the signal by 2.3 samples. In a parallel sum with Drive this is not compensated.

5.5 Filter crossing guard v18

Settings → Community features → **Filter crossing guard** (7-segment CR05), off by default. When the HPF and the LPF come close or cross, it lowers both resonances smoothly, so their peaks don't add up to a painful jump in level.

HPF position	Combined peak, with the guard	Without
less than half an octave below the LPF, or above it	at most 0.7 dB above the larger single peak	up to 39 dB
half an octave to an octave and a half below	at most 1.7 dB	

- It starts acting from two octaves apart. A self-oscillating ladder stops singing at the crossing.
- Less resonance raises the passband slightly. In single cases (LP12 with the SVF as HPF) the overall level can rise by up to 3 dB.

5.6 EQ with real shelves v18

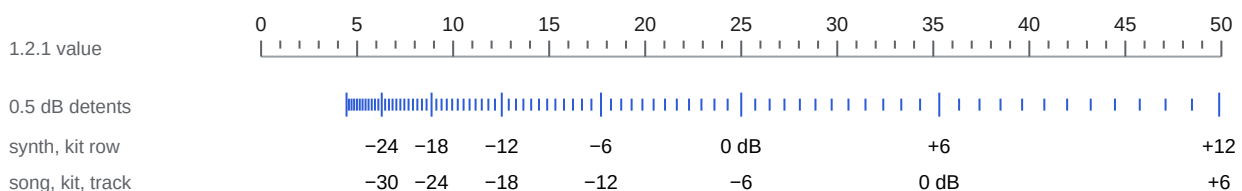
- Bass and treble are now real shelving filters instead of one-pole crossovers. The knobs and the corner frequencies stay: 199 Hz and 2.8 kHz.
- The noise floor drops from -77 to -141 dBFS. Clicks when you turn drop from -24 to -70 dBc.
- Set to neutral, the EQ passes the signal through bit for bit.
- With both bands in use it needs twice as much CPU as before.

5.7 Volume in dB v18

- **Every detent is 0.5 dB**, in the menus and on the gold knobs. The display shows dB: "-3.5 dB" on the OLED, -3.5 on the 7-segment display, OFF at the very bottom.
- **Songs are saved as before.** They sound equally loud and still load in v17. Their values simply lie between the old positions 0–50.



The song's volume knob: master level -8.5 dB.



Where the old values 0–50 lie in dB. Each blue tick is one detent of 0.5 dB, drawn from -30 dB up. Low down, one old step spans several dB; high up it spans less than one.

- **One ramp for everything:** sound, kit row, kit, audio track and song ramp volume × pan × sidechain × filter compensation per sample, exactly onto the target. A jump of more than 6 dB takes at least 5 ms, also from automation and when several parameters change at once.
- **So the sidechain also ducks over at least 5 ms.** Very hard pumping effects in old songs sound a bit softer.
- **More precise:** quiet settings no longer lose resolution. The level before the filters stays as in v17, so turning up doesn't change how much they saturate.

Old value	20	25	30	35	40	45	50
Synth, kit row (default 40)	-3.9	0.0	+3.2	+5.8	+8.2	+10.2	+12.0
Song, kit, audio track (default 35)	-9.9	-6.0	-2.9	-0.2	+2.1	+4.2	+6.0

In dB as the display shows it. From one value a to another b the level changes by $40 \cdot \log_{10}(b/a)$ dB. A kit also has a fixed +1.9 dB as compensation for the filter resonance.

5.8 Levels, headroom and the output limiter

No volume control in the Deluge distorts by itself: each is a pure multiplication in 32 bits with over 30 dB of headroom. **Only the sum at the output clips**, hard at 0 dBFS, the same on headphones, line out, USB and resampling. You hear it first on the track you are turning up.

- **The VU meter shows the mean, not the peaks.** It can already clip while the meter shows yellow. The only official target: not above the 3rd pad from the top (-4.5).
- **A baseline** from the level analysis of real songs; there is no official one:

What	Suggested value
Song volume	35 to start, then down until the peak lies at -6 dBFS. Turning down in 32 bits costs no quality.
Kit, audio track	At most 35.
Synth, kit row	40 as the upper limit for loud things like kick and bass; 45–50 only for quiet samples.
Master compressor	Off. Switched on, it makes up its reduction with make-up gain, so it creates no headroom.
Peak	At most -6 dBFS. In DelugeRec the pad at -3 stays dark (§8.2).
VOLUME knob	Any. It is analog, after the converter, and doesn't act on USB and resampling.

DelugeBaseline checks all songs of a card against this baseline and can set them to it (§8.4).

Output limiter v18: Settings → Community features → **Output limiter** (7-segment LIMT), off by default. A peak limiter that looks ahead 0.7 ms and sets in softly 1 dB below full scale, without aliasing. It acts only where the output would otherwise clip hard.

- When it is on, the output comes 0.7 ms later. When you switch it on or off, the output jumps once by this time, which can click briefly.
- When it is off, everything stays bit for bit the same.

5.9 Sidechain v3, v18

- **No click when ducking.** In 1.2.1 the volume ramp of each buffer started at the new value and overshoot; with more than 6 dB of ducking within one buffer, the gain even flipped polarity. Now the ramp runs from the old to the new value, and the reverb return follows smoothly. The bug is also in the current community firmware.
- **At least 5 ms** for jumps over 6 dB since v18 (§5.7).
- **The song's reverb sidechain keeps its sync level** v18.3. Since 1.2.1 it slipped one step lower at every save and load. Existing songs load with the saved value.

5.10 Reverb v10, v14

All settings are in the reverb menu as before, for the song and in every sound.

Setting	Models	Values and effect
Model → Digital (DIGI)	new	A plate after Jon Dattorro (1997), built like the Lexicon 224: dense and smooth, without metallic ringing. As loud as Mutable (within 0.5 dB), so switching the model doesn't jump in volume.
Room size	Digital	The time: 0 ≈ 0.6 s, 30 (default) ≈ 4.5 s, 45 ≈ 18 s, 50 almost endless.
Width	Digital	Stereo width, 0 = mono.
Damping	Mutable, Digital	0 bright, 50 dark, stepless. On Mutable it ran backwards in 1.2.1. Songs sound the same; only the number shown is now 50 minus the old one (default 36 → 14).
HPF	Mutable, Digital	Repaired: 0 = 20 Hz, 25 = 190 Hz, 50 = 540 Hz, saved and loaded with the song. Old songs load with the frequency they really had (old 50 → new 12).
LPF	Mutable, Digital	New: 0 = 500 Hz, 25 = 3.2 kHz, 49 = 18.6 kHz, 50 = off (default). It darkens the whole reverb; damping darkens the tail more and more over time.
Modulation v14	Mutable, Digital	0–50, default 0. At 0 the reverb stands still: a held tone stays steady in it. 50 sounds like v10 to v13, with the slow movement that fluctuated up to about 16 dB in level and some cents in pitch.
Pre-delay v14	all	0–100 ms, default 0. The reverb starts later, and the dry sound stands on its own first.

- **Mutable** at Modulation 50 moves at the speed of the original Mutable Instruments modules (about 0.45 and 0.28 Hz). 1.2.1 ran it 16 times slower, almost static.
- **Freeverb**: with width below the maximum the right channel was louder (by 3.6 dB at width 0); now balanced. A long, loud reverb no longer wraps around with a loud click. Freeverb has no HPF, LPF or modulation.

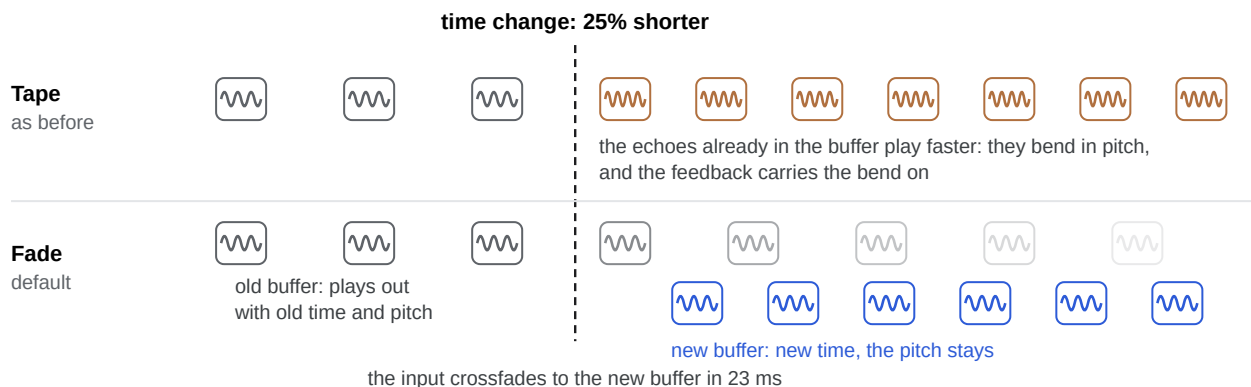
Against muddiness: 10–30 ms of pre-delay make the reverb much clearer without sounding detached. HPF at about 25 (190 Hz) takes the bass out of the reverb. A bit more damping darkens the tail. Without modulation the highs ring a bit longer on Mutable; raise damping a few steps if that is too bright.

- **Old songs sound stiller:** songs without these settings load with Modulation 0 and no pre-delay. Set Modulation to 50 for the old sound.
- Digital needs about a third more CPU than Mutable, computed once for the whole song.
- A song saved with Digital plays without reverb in 1.2.1.

5.11 Delay **v11, v14**

For sounds, kits, audio tracks and the song. Two new menu items:

- **Time change** (after Type) **v14**: **Fade** (default) or **Tape**.
- **LPF** and **HPF** in the feedback (after Sync) **v11**: each repeat gets darker or thinner than the one before, as with tape and bucket-brigade delays. LPF: 0 = 500 Hz, 25 = 3.2 kHz, 49 = 18.6 kHz, 50 = off (default). HPF: 0 = off (default), 1 = 25 Hz, 25 = 190 Hz, 50 = 540 Hz. They are saved only when switched on.



What happens to the echoes when the delay time gets shorter. With Tape they bend in pitch, as before. With Fade the old echoes play out unchanged while the new time starts in a fresh buffer; the pitch stays accurate to 0.01 cents.

- **Old songs:** delays with a modulated or automated delay rate load as Tape, so they sound as before. All others load as Fade.
- **Clean repeats** [v11](#): in 1.2.1 the delay could stay in its resampling mode for good after a time change, even if you only moved the knob briefly and back. Every repeat then lost 9 dB at 10 kHz. Now the repeats are bit-exact again after a change.
- **While the time moves** (LFO, envelope, automation) the delay resamples with a cubic kernel: a repeat loses 0.6 dB instead of 6.9 dB at 10 kHz, and the resampling noise drops from -59 to -80 dB.
- **No clicks** when the time changes, and no click on the first echo after a pause. The end of the echoes fades out instead of cutting off.
- **Up to 4 s** of delay without aliasing (buffer up to 4 s instead of 2 s).
- **Analog mode** saturates with less aliasing.
- **Fixed from 1.2.1:** below 3% feedback the delay threw its buffer away on every round. A copied sound keeps triplet or dotted in its delay sync.

The delay is 1 sample (0.02 ms) longer than in 1.2.1, because all its write paths now use the same position.

5.12 Try it by ear

v18.x has not been played on a real Deluge yet. These short checks, ideally with headphones, show the changes. Please report anything that sounds wrong (§11.8).

Check	What to do and what to expect
Rustle	A synth with LPF 24 dB, cutoff low (about 10–20), resonance medium, a long note. The band of noise around the note is gone. The same with 12 dB and Drive.
Transitions	Turn cutoff and resonance quickly; switch the filter mode and routing while a note sounds. No steps, no clicks.
Drive	LPF mode Drive with a lot of resonance: the bass stays. A song with several Drive filters doesn't lose voices at the start of the bar.
EQ	Turn bass and treble while music plays. No clicks.
Volume	Every detent changes 0.5 dB, the display shows dB, at the very bottom OFF. Tracks set quiet sound clean.
Output limiter	On a song that clips: no more hard distortion. Switching it may click once.
Crossing guard	Turn HPF and LPF with resonance towards each other: no painful jump in level.
After a silence	Let a kit pause (a gap in the clip, or muted), change its LPF, EQ or volume meanwhile, let it play again. The first hit comes without a tick or a bright attack, right away with the new sound.

6. Arpeggiator

mastertune has the complete arpeggiator of community firmware 1.3, with the menus in the familiar 1.2.1 style: one list, no horizontal menu. New items sit where they belong by topic. On top come latch, ratchet bounce and the ping-pong roll, which 1.3 doesn't have. [v5](#) [v6](#) [v13](#)

6.1 What's new

Feature	Where	What it does
Kit arpeggiator	Kit with AFFECT ENTIRE pressed → menu → Kit arpeggiator	Plays a kit's rows like the notes of a chord (Up, Down, Random, Walk, Pattern ...). Switch it off per row with "Include in kit arp".
Arp for MIDI and gate rows	Select the row → menu	Its own arpeggiator and randomizer, with the shortcut column 11 as on MIDI tracks. Before, there was only a warning LED.
Preset, with Walk	Arpeggiator → Preset	Now also in the list, not only on the pad.
Latch new	Arpeggiator → Latch	The arp keeps playing after you let go (§6.2).
Step repeat	Arpeggiator → Step repeat	Each step is repeated 1–8 times.
Note modes	Arpeggiator → Note mode	Walk 1–3: randomly one step forward or back. Pattern: a random but repeating order; select it again to roll a new one.
Chord simulator	Kit row → Arpeggiator	A drum row plays a chord: 5th, sus2, minor, major, sus4, m7, 7, maj7.
Ratchets and bounce new	Arpeggiator → Ratchet notes, Ratchet bounce, Bounce length, Bounce velocity	Ratchet hits that speed up or slow down like a bouncing ball, up to a ping-pong roll (§6.3).
Randomizer	Its own menu right after Arpeggiator	Lock, spreads and probabilities (§6.5).
Automation	Automation view, SELECT encoder	All new arp parameters of synths and kit rows; for the kit (Affect Entire) also those of the kit arp.

"new" marks what community firmware 1.3 doesn't have either.

6.2 Latch

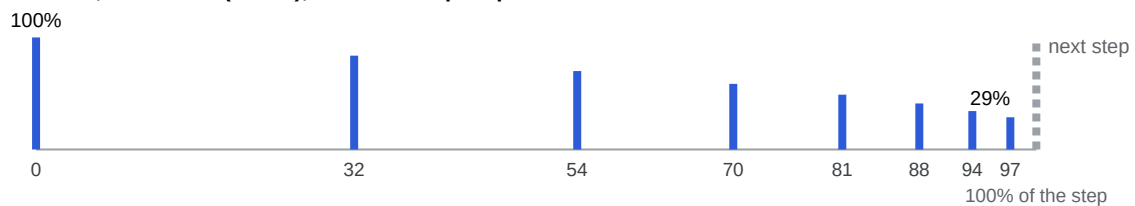
With **Arpeggiator** → **Latch** on, the arp keeps playing after you let go of the keys. The next note you play after releasing all keys replaces the held notes, without the arp falling out of time. Switching Latch off stops the held notes.

6.3 Ratchets, bounce and roll

Setting	Values	What it does
Ratchet notes v6	Auto, 2 ... 8, Roll	Auto , as in 1.3: randomly 2, 4 or 8 hits, weighted by Number of ratchets. 2 ... 8 : always exactly that many; whether a step ratchets is then decided only by Ratchet probability. Roll v13 : as many hits as it takes until the span is used up (below).
Ratchet bounce	-10 ... +10	The hits of a ratchet like a bouncing ball. Each gap is r times the one before: $r = 0.95$ at +1, 0.8 at +4, 0.7 at +6, 0.5 at +10. Positive values get faster, negative ones slower. 0 = even, as before.
Bounce length v13	1 ... 16	Over how many arp steps one ratchet lasts. The note holds, then the arp continues with the next note. Rhythm and sequence length stay in time.
Bounce velocity v13	Even, Fade, Rise	Even : all hits equally loud. Fade (default): the faster, the quieter, like a ball. Rise : the faster, the louder, like ping-pong. It was called "Bounce fade" (on/off) in v6.
Ratchet probability	0 ... 100%	How often a step ratchets.

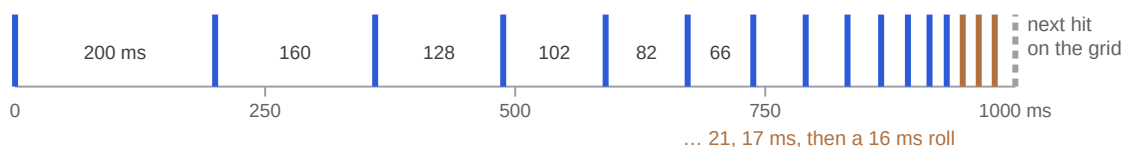
- **Short syncs**: with a fixed number of ratchet notes, sync 1/128 and 1/256 allow at most 2, sync 1/64 at most 4. Otherwise the gaps would hardly be longer than an audio block (2.9 ms).
- **Even ratchets** divide the step by the number of notes. For 2, 4 and 8 that gives exactly the same times as before.
- **Compatible**: with the defaults (Auto, Fade), Bounce length 1 and without Roll, the arp plays exactly as in v5 and v12.

Ratchet notes 8, bounce +6 ($r = 0.7$), Fade: one arp step



Roll, bounce +4 ($r = 0.8$), Bounce length 4, 120 BPM: the preset PINGPONG ROLL

Hit j falls at $S \cdot (1 - r)^j$ of the span S . The gaps shrink by r each time and converge on the end of the span.



Top: a fixed number of hits in one step, each gap 0.7 times the one before, getting quieter from 100% to 29%. Bottom: Roll over four eighths. Once a gap would fall below 16 ms, the hits continue every 16 ms (brown) up to the next hit on the grid: 16 hits in all.

Negative bounce plays the same pattern backwards: the hits start dense and spread out.

6.4 The ping-pong presets

Two synth presets show the roll. Copy PINGPONG ROLL.XML and PETTRA PINGPONG.XML from mastertune-1.2.1/presets/SYNTHS/ into the SYNTHS folder of your card. Load them in a synth clip like any other preset; the arp settings come with them. They need v13 or later.

Setting	PINGPONG ROLL	PETTRA PINGPONG
What it is	A ping-pong ball between two paddles that close in: per ratchet 16 hits over 4 eighths, ever faster and louder	The figure from Pettra, "You Are The Seeds": 3 equally loud hits in one eighth, each gap $\times 0.7$
Sync	1/8	1/8
Ratchet notes	Roll	3
Ratchet bounce	+4	+6
Bounce length	4	1
Bounce velocity	Rise	Even
Ratchet probability	100%	100%

The presets store no tempo. Pettra runs at 138 BPM; there the three hits fall 0, 99 and 169 ms after the start of the eighth.

- **A fill only now and then:** lower the Ratchet probability. Or, in automation view, set it only on the steps before which the fill should come, and 0 elsewhere.
- **Longer or shorter:** change Bounce length or Sync. At 1/16 and Bounce length 8 the fill also lasts half a note.
- **Softer or harder acceleration:** Ratchet bounce +1 ... +10.
- **The hit after the fill should bang:** set Ratchet probability below 100%. The landing step is then a normal note at full velocity. At 100% the next fill starts right there, and Rise starts it at half velocity.

6.5 Randomizer

The randomizer has its own menu right after Arpeggiator:

- **Lock:** repeatable randomness over 16 steps.
- **Spreads:** gate, octave and velocity.
- **Chord** polyphony and probability.
- **Probabilities** for note, swap, bass, glide and reverse.

Reverse probability plays single arp notes with their sample backwards. Unlike in 1.3 this applies per voice, so notes sounding at the same time don't reverse each other.

6.6 Songs, presets and compatibility

- **1.2.1 songs and presets** load unchanged, including MIDI and CV tracks with arp settings.
- **Stock 1.2.1** loads mastertune files but skips the new entries (kit arp, randomizer, latch, bounce) and loses them at the next save. Walk and Pattern become Up there.
- **Automation view:** the internal parameter numbers now match those of 1.3. This only changes which parameter is preselected when you first open an old song in automation view. Values, automations, mod knob and MIDI learn assignments are saved by name and are not affected.
- **Same start every time:** in 1.2.1 LFOs, mod FX and the arp started at a random point after loading. Now they always start the same way. v13
- **deluge-editor** doesn't show the arp additions (§8.7). Set them on the Deluge; they are kept in the file when the editor saves.

Not taken over from 1.3: the pad shortcuts in column 15 (velocity spread, Lock, Note probability), because 1.2.1 uses these pads for patching; and the MIDI Follow CCs for arp parameters, because they would have shifted existing assignments.

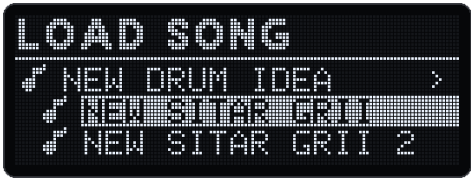
7. Songs and workflow

A song browser that keeps a song's versions together, a countdown when you change songs, saving without crackle, a dimmer OLED and pads that don't flicker, plus fixes for MIDI, clock and the card.

7.1 Song browser with versions v17, v18

Songs whose names differ only by a number at the end are versions of one song: "New Sitar Grii", "New Sitar Grii 2" ... "New Sitar Grii 10" appear as **one line with an arrow**. The line shows the whole name.

A number after "-" or "." belongs to the name: "TR-808" and "Jam 2026-09-27" stay songs of their own.



The versions of "New Sitar Grii", folded out and indented. A line with an arrow is a group that is still closed.

To ...	do this
open a group	Click: press <code>SELECT</code> or <code>LOAD</code> . The versions appear indented below it.
load a version	Turn to it and click. This also works during playback, with the countdown (\$7.2).
close a group	<code>BACK</code> on one of its versions. Groups you opened stay open while you keep turning, up to 8.
find the loaded song	Open the browser: the group of the loaded song is already open.
load by name	Type the name; it loads directly.
delete	<code>SHIFT</code> + <code>SAVE</code> on a version. On a closed group line it is blocked: "Open the group first".

Saving and the other browsers are unchanged. The browser reads the folder less often than before, and while you browse during playback the audio routine pauses for at most 0.39 ms.

7.2 Song change with a countdown v14

Load a song while another one is playing, and the Deluge counts down to the change:

- as long as repeats remain, it counts the **loops**;
- in the last loop the **bars**;
- in the last bar the **beats**, 4-3-2-1.

It stays right with swing and with an external MIDI clock.



Song change armed: 2 bars remaining. The list with the next song stays visible.

- **OLED:** the countdown stays in the title line, for example "Bars remaining 3".
- **7-segment display:** the number, blinking while it counts loops. Beats carry a dot, so you can tell them from bars.
- **Until the change** the gold knobs and mod buttons control the master effects of the **playing** song, from the moment you press `LOAD`, even with Affect Entire off. The new song keeps its own values.

The countdown has priority v19.0.2. While a countdown runs, nothing covers it:

- **OLED:** it stays on the title line, drawn over everything else. A popup, such as a gold knob's value, the filter type or the DJ filter, appears below it.
- **In Song view too:** when clips or a section are armed to launch, the countdown (the bars until the launch, in the last bar the beats) is on the title line. Before, it was a box in the middle of the screen, which every knob covered; now the song's name stays visible below it.
- The CPU monitor's line moves to the bottom while a countdown takes the top.
- **7-segment display:** the gold knobs and mod buttons show no popup while it shows a countdown. They still do what they do.



Song change armed, the DJ filter turned: its popup below the countdown.



Song view, clips armed to stop: the countdown on the title line, a gold knob's value below it.

7.3 Saving without crackle v11, v13

- **No crackle:** in 1.2.1 the audio engine stood still while the Deluge wrote a song or preset to the card. The output repeated its last buffer: a short crackle at every save. Now the audio keeps running, and the samples that are playing keep loading. This applies to songs, synth and kit presets and settings.
- **Faster:** saving during playback takes 22 instead of 344 ms in the emulator (a song with 3 synths), or 66 instead of 716 ms (5 synths).
- **No loading window while you browse:** a small animation runs at the top right of the OLED while loading and saving, as in community firmware 1.3. The preset name stays visible.

7.4 OLED brightness v18

Settings → **OLED brightness**, levels 1–10. 10 is as bright as before, 1 is still readable. Each turn takes effect at once, and the level applies again after a restart. The readable firmware version is described in §1.4.

7.5 Flicker-free pad dimming v13

Settings → **Community features** → **Flicker-free dimming**, on by default.

- **Why the pads flickered:** 1.2.1 dims the pads through the LED controller with dark pauses. Below 40% brightness the pauses get so long that the whole refresh slows down, 6.8 times at 0%.
- **Now:** the LED controller always refreshes at full speed and dims only down to 43.5%. The firmware dims the rest through the colours of the pads, the sidebar and the gold knob LEDs. Every level gives the same amount of light as before.
- **Trade-off:** the button LEDs, the 7-segment display and pads that the LED controller blinks by itself (the fast play cursor, blinking shortcuts in menus) dim only down to 43.5%. At very low brightness they are brighter than before.
- **Off** sends byte for byte the same as 1.2.1.

7.6 Settings that survive v18.3

When the Deluge saves its community settings, it now keeps those it doesn't know, for example from a newer version, with their name and value. 1.2.1, v17 and the current community firmware wrote them back under a wrong

name. So switching to v17 and back still resets Output limiter, Filter crossing guard and OLED brightness to their defaults. Set them again after such a switch.

7.7 MIDI, clock and card fixes

Fix	Since	What changed
MIDI and gate outputs on time	v13	Events late in an audio block went out a whole block (2.9 ms) too early, especially while saving. And the timer of the MIDI and gate outputs was never reset before the start: a run could come up to 38 samples early, or after an overflow about 127 ms late. Now every run is accurate to ± 1.4 samples.
Late clock bytes	v17	A clock byte that is read late now ticks at once instead of up to 2.9 ms late.
Clock outputs under external clock	v15	When the Deluge followed an external MIDI clock, its gate clock and MIDI clock outputs stopped after a short burst. Now both keep running in time with the incoming clock. With the internal clock everything stays as before.
Section launch by CC	v15	A section launch learned to a CC was lost when the song was loaded; learned notes were kept. Now the CC is kept too.
USB MIDI from the computer	v15	If MIDI came faster than once per millisecond (dense automation from the DAW, SysEx, file access over USB), packets were lost. Now the computer waits until the Deluge is ready.
Card transfers	v14	Around every transfer from and to the SD card, a write happening at the same time right next to the buffer could get lost. This bug is from 1.2.1.
File access over USB	v14	In rare cases wrong bytes could end up in a copied file (§8.7).

The three fixes of v15 were taken over from the current community firmware.

8. Tools for the computer

Three small Windows programs record the Deluge, retune a sample library and even out the levels of a card. Two browser pages log the CPU monitor and the profiler, and two web apps reach the SD card over USB.

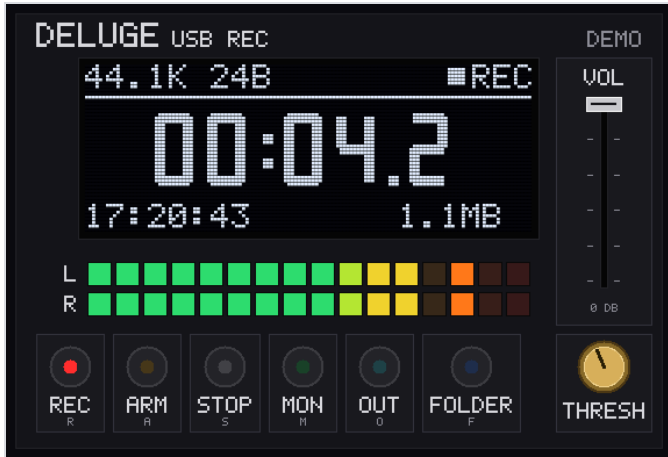
8.1 Overview

Tool	What it does	Where to get it
DelugeRec	Records the Deluge's USB audio, with a monitor on the headphones (§8.2).	Release <code>deluge-rec</code> (v7)
DelugeTuner	Retunes a card's sample library once to the master tune (§8.3).	Release <code>deluge-tuner</code> (v2)
DelugeBaseline	Checks the levels of all songs on a card and evens them out, with a backup (§8.4).	Release <code>deluge-baseline</code> (v4)
CPU monitor page	Logs the CPU monitor live and as CSV (§8.5).	<code>tools/cpu_monitor.html</code>
Profiler	Shows the CPU time per track, task and function (§8.6).	<code>tools/profiler.html</code> , <code>tools/deluge_profiler.py</code>
DEx, <code>deluge-editor</code>	Files and presets on the SD card over USB (§8.7).	<code>dex.silicak.es</code> , <code>cyface.github.io/deluge-editor</code>

- **The Windows programs** are one `.exe` each, with nothing to install. The newest version is always in its release at `github.com/Giansn/deluge/releases`. Windows asks once, because they are not signed: click **More info**, then **Run anyway**.
- **As Python scripts** they are in `mastertune-1.2.1/tools/`.
- **All three** have the Deluge's look, and each button shows its keyboard key below it. DelugeTuner and DelugeBaseline switch between German and English (`SPRACHE / LANGUAGE`).

8.2 DelugeRec

DelugeRec records the "Deluge" input from USB audio (chapter 4) bit-exact as a 24-bit WAV file. On Windows it uses WASAPI exclusive mode, and the display shows whether 24 bits really arrive. It only listens: it sends nothing to the Deluge.

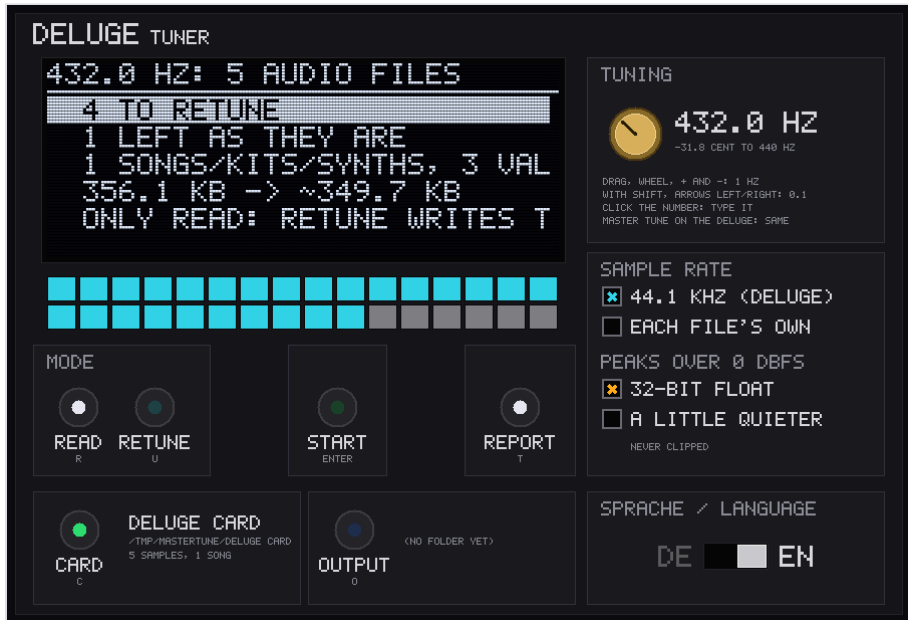


DelugeRec while recording: 44.1 kHz and 24 bit arrive, the pads show the level.

Control	What it does
REC	Starts recording.
ARM	Starts as soon as the signal exceeds a threshold (THRESH), with 0.3 s of pre-roll.
STOP	Ends the take.
MON	The monitor on the headphones.
OUT	Chooses the monitor's output: up/down, mouse wheel or the knob, then Enter. The Deluge itself is never offered.
FOLDER	Opens the folder with the recordings.
VOL	Leave it at 0 dB: then the recording is bit-exact. Turning it down doesn't repair clipping that already happens in the Deluge.
Pads	The level. For the baseline in §5.8 the pad at -3 should stay dark.

- **Take names:** with firmware v18 or later, DelugeRec names each take after the song, the date and the firmware, for example "Song name, 28.09.2026 - 1.2.1 v18.3.WAV". The Deluge reports the song name while the audio stream is open (§4.5). From v18.2 on you need DelugeRec v7: v6 cuts the version off at the dot.
- **As a script:** `pip install sounddevice numpy`, then `python tools/deluge_rec.py`. Without a Deluge: --demo.

8.3 DelugeTuner



DelugeTuner after READ: what it would retune at 432.0 Hz, and how much.

Away from 440 Hz the Deluge resamples every sample voice (§2.5). DelugeTuner converts the sample library of a card once to the master tune. The converted files carry the `mtun` chunk (§2.3), so the Deluge plays them directly again. The firmware needs no change for this.

1. **CARD:** choose the SD card or a copy of it. It is only read.

2. **OUTPUT:** choose a folder on the computer for the new card. An empty folder becomes the new card; otherwise a new folder "Deluge 432 Hz" is created inside it, numbered if needed. A folder inside the card is refused.
3. **Gold knob:** the tuning, 415.3 to 466.2 Hz, 432 Hz at the start. Drag, the mouse wheel or + and – change it by 1 Hz; with Shift or the arrow keys left and right by 0.1 Hz. Click the number to type it.
4. **Checkboxes:** 44.1 kHz or each file's own rate; peaks above 0 dBFS as 32-bit float or a little quieter.
5. **MODE:** **READ** only shows what it would do, **RETUNE** writes the new card. Press **START**; display and pads show the progress, then the summary. **REPORT** opens RETUNE_REPORT.txt (in English).
6. Copy the new card to an empty SD card and set the same master tune on the Deluge.

Beforehand it checks the free space at the output folder: about 5% more than the card. **ESC** or closing the window stops as soon as the files in progress are finished. **START** with the same card, output folder and settings continues where it stopped.

What it does to the card:

- Resamples every sample at the exact ratio ($440/432 = 55/54$): a 440 Hz tone then measures 432.00 Hz, and the file gets 55/54 longer. 48 and 96 kHz files become 44.1 kHz in the same pass.
- Adjusts start and end markers, loops and audio clip positions in all songs, kits and synths, and the `smpl` and `cue` chunks in the files.
- Gives time-stretched audio clips and samples a `_ts` copy that only shifts the pitch and keeps the length. AIFF files become WAV, because the Deluge reads `mtun` only in WAV.
- Clips nothing: a file whose peaks would go above 0 dBFS is written as 32-bit float, or made just as much quieter as needed. The Deluge limits float samples to 0 dBFS when it loads them. The report lists these files.
- Never changes the original card. A second run changes nothing more. Wavetables, files that already carry the target tuning and files the Deluge can't read stay as they are.

What it gains (emulator, 432 Hz): every untransposed voice plays directly. Per voice and block the work drops from 13,300 to 2,490 instructions (mono) and from 19,800 to 3,000 (audio clip). The whole test card needed 34% less without the sample cache, 6% less with it.

- **Always the whole card:** convert a copy of the whole card and copy the whole result to a second card, never single folders. Otherwise samples that several songs share would be converted, but the positions in the other songs would not.
- **Transposed notes** are still resampled, for example a synth sample played melodically. At 440 Hz the converted samples sound right thanks to `mtun`, but cost resampling again.
- **Very short loops and single-cycle samples** can be detuned by up to 1.4 cents, because loop lengths are rounded to whole samples. The tool warns about this. In `_ts` copies attacks can shift by up to 4.6 ms.
- **Loops that no song uses** are treated like single samples. If you later make an audio clip from one, the Deluge stretches it again. Take such loops from a copy of the originals.
- Converted cards have not been played on a real Deluge yet.

Command line: `pip install numpy soxr pylibrb`, then `python3 tools/retune_library.py --card CARD_COPY -out NEW_CARD` (default 432 Hz, 44.1 kHz). `--dry-run` only shows, `--resume` continues after an interruption, `--no-float` makes peaks quieter instead of writing float.

8.4 DelugeBaseline



DelugeBaseline after reading a card: one pad per song, green in order, orange with notes.

DelugeBaseline checks all songs on a card against the level baseline in §5.8 and can set them to it. It has two functions, each in the mode **READ** (only shows) or **APPLY** (shows, then writes on the second START):

Function	What APPLY changes
LEVELS	Song volume above 35 to 35, master compressor off. Kits and audio tracks above 35 to 35. Synths and kit rows down far enough that they act at most like 40 with a fully driven sample. Automation keeps its shape. Saturation, the tracks' compressors, analog delay and filters stay: that is sound, not level.
NORM	Raises every sample under SAMPLES/ whose peak lies below the target (TARGET knob, default -1 dBFS) up to the target, never above. Nothing clips; format, bit depth and chunks stay. With COMPENSATE (on by default) every oscillator that plays a raised sample is turned down by exactly as much, in every song, kit and synth, so the songs sound as before. Wavetables and audio clips are never touched.

- **Buttons:** FUNCTION LEVELS (L) or NORM (N), MODE READ (R) or APPLY (A), START (Enter), CARD (C), REPORT (T) opens the whole report as text, RESTORE (B) shows the backups.
- **APPLY** first shows what would change, and the pads of the songs concerned blink. START again writes, Esc cancels.
- **WRITE TO:** SD CARD DIRECTLY, with the old files in BASELINE-BACKUP/, or COPY FOLDER, with only the changed files in the card's layout. RESTORE plays a backup back.
- **The report** lists songs, kits or audio tracks above 35, a master compressor that is on, synths and kit rows above 40 (marked "no notes" if the row has no notes), and stages that distort more with level: saturation, compressor, analog delay with feedback, low-pass with drive, filters with resonance from 25. Every level is shown as the value and in dB as on the Deluge, for example "40.0 (+8.16 dB)".
- **Its limit:** it doesn't tell you whether a song clips. Only measuring shows that, for example with DelugeRec.

Without a window: `py tools/deluge_baseline.py [--lang en] check|levels|normalize E:\ [--yes]`. Without -yes it only shows what it would do.

8.5 The CPU monitor page

1. Connect the Deluge by USB and switch the CPU monitor on (§3.4).

2. Open `tools/cpu_monitor.html` in Chrome or Edge; dragging the file into the window is enough. Click **Connect** and allow MIDI with SysEx.
3. Leave the input on "All inputs" or choose the Deluge's port 3. Tiles and a graph of the last 5 minutes run live.
4. **Export CSV** saves every second received: one line per second, semicolon as separator.

Value	Meaning
CPU mean	Computing time of the audio routine divided by the duration of the audio it produced. Above about 90% all the time, it gets tight.
CPU peak	The most expensive single pass. Single peaks above 100% are absorbed by the output buffer (128 samples, 2.9 ms).
Voices	Active voices, now and the maximum.
Direness	0–14. Above 0 the audio routine is behind the output and the quality is lowered (QL). The share says how much of the audio was made that way.
Culling	Voices per second that the CPU protection really fades or switches off (VC). Voice stealing because of the polyphony doesn't count.
SD card	Pure read time of a sample cluster, mean and longest. It shows how fast the card is.
SD latency incl. audio	The same accesses from start to end, including the audio computed meanwhile: how long streaming waits for a cluster. If it is much bigger than the card time, the CPU slows streaming down, not the card.
Longest audio gap	The longest interval between two passes of the audio routine. From about 2.9 ms on, dropouts threaten.

8.6 The profiler

With **Settings** → **CPU monitor** → **Profile** the Deluge shows the usual line and also sends, 1000 times a second, where in its code it is working, plus the exact CPU time of each track. v16 It costs about 0.04% CPU and about 8 KB/s on USB MIDI port 3; the buffer always leaves 1 KB free, so notes and clock don't wait.

1. Open `tools/profiler.html` in Chrome or Edge and click **Connect**.
2. **Load symbols:** choose the `.symbols.json` next to the firmware you installed. Only the matching file gives the right function names.
3. Set the Deluge to Profile and play. **Reset** separates two parts, for example idle and playing.
4. Read the tables **by track**, **by task** and **by function**, or **Save recording** as a `.jsonl` file.

While a track computes, interrupts are off. That time shows up collected as `Song : : renderAudio`; which track it was is under "by track".

Without a browser, `tools/deluge_profiler.py` does the same (`pip install mido python-rtmidi`; on Windows use Python 3.12). `ports` lists the Deluge's three ports, live prints one line per second, `record -s 60 -o profile.jsonl` records, `report profile.jsonl --symbols FILE` evaluates. It finds port 3 by itself. The live line:

```
CPU 43.0 % (peak 71.0) voices 24 (max 30) QL 0 (0.0 % of the time) cut 0 gap 3.1 ms SD 11 loads, avg 1.7 max 9.0 ms
```

Part	Meaning
CPU	Load of the audio windows: mean and peak of the last second. From v17 on this is the real load; up to v16 an idle Deluge showed 85–90% of busy waiting.
voices	Sounding voices, now and the maximum.
QL	Quality lowered: the direness (0–14) and for how much of the time.
cut	Voices cut by the CPU protection. The number that counts.
gap	The longest pause between two runs of the audio routine. From about 2.9 ms on a dropout threatens.
SD	Loads from the card, mean and longest duration.

With Profile, Live adds every 5 s the audio routine's share, the most expensive tracks and tasks and, with - - symbols, functions.

8.7 The SD card over USB: DEx and deluge-editor v7

The card stays in the Deluge. Connect the Deluge by USB, open the web app in Chrome, Edge or Opera and allow MIDI access. The apps find the file access by themselves.

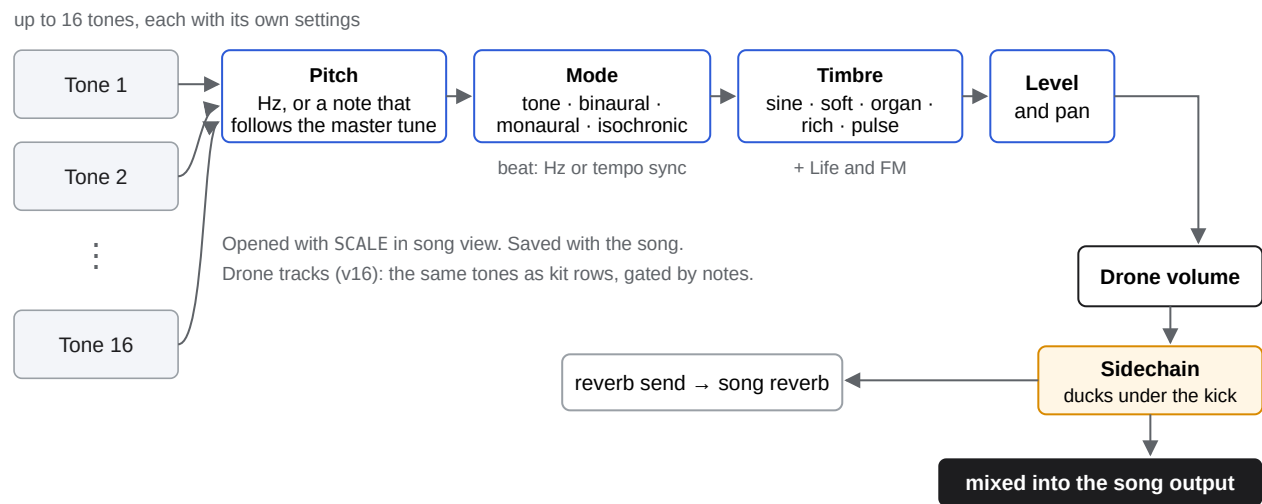
App	What works
DEx dex.silicak.es	File browser: view folders, upload and download files, rename, copy, move, delete, create folders. As before also display mirroring and screenshots.
deluge-editor cyface.github.io/deluge-editor	Open synth and kit presets directly from the card and save them back.

File access works only over USB; the Deluge ignores such requests on the DIN sockets. Its USB MIDI send buffer is four times as large as in 1.2.1, so even a full folder page with long names fits.

- **deluge-editor shows "needs community 1.3.0 or later" in red**, because the Deluge honestly reports 1.2.1. Ignore it: opening and saving work.
- **deluge-editor hides controls it only knows from 1.3 on**, among them the arp additions (chapter 6). Set those on the Deluge; they are kept in the file. "Live Edit" doesn't work.
- **Names with umlauts** are shown wrongly, and such files usually can't be opened. Use only A–Z, 0–9 and _ in names.
- **During playback** don't delete or overwrite samples the song needs. Large transfers are slow over MIDI; for whole sample collections a card reader is faster.
- **Don't transfer files while you play with a DAW over USB**: notes and clock to the computer share the send buffer and can arrive late. DIN MIDI is not affected.

9. The frequency drone

The drone plays up to **16 sustained tones** under the music of your song. Each tone can be steady or carry a beat: **binaural**, **monaural** or **isochronic**, free in hertz or synced to the tempo, also in triplets. You set the pitch in hertz or as a note that follows the master tune. Isochronic pulses have attack and release. The drone ducks under the song's sidechain, can send to the song reverb and is saved with the song. With drone tracks, its tones also live in kits, so clips and the arranger can switch and automate them.



How the drone is built: every tone has its pitch, mode, timbre, level and pan; the whole drone has a volume, ducks under the song's sidechain and can send to the song reverb.

Version	What the drone got
v12	The drone: up to 16 tones, binaural, monaural and isochronic beats, tempo sync, sidechain, its own view
v13	Select sets the pitch in detents; the pads light steadily; attack and release for isochronic pulses; triplets; gold knobs by mod section; reverb send
v15	Life (slow drift), FM and the Pulse timbre
v16	Drone tracks: drone tones as kit rows in song and arranger view, a Hz track per row (section 9.16)
v17	Fix: in v13–v16 the Deluge could hang about 15 ms after the drone view opened. Use v17 or later.

Start quiet. Sustained tones, especially low ones and many at once, can seem louder than expected on headphones. Start at a low volume and raise it slowly.

Not a therapy: The names Delta to Gamma only describe frequency ranges. How beats affect people is scientifically disputed. The drone is a sound tool, not a medical device.

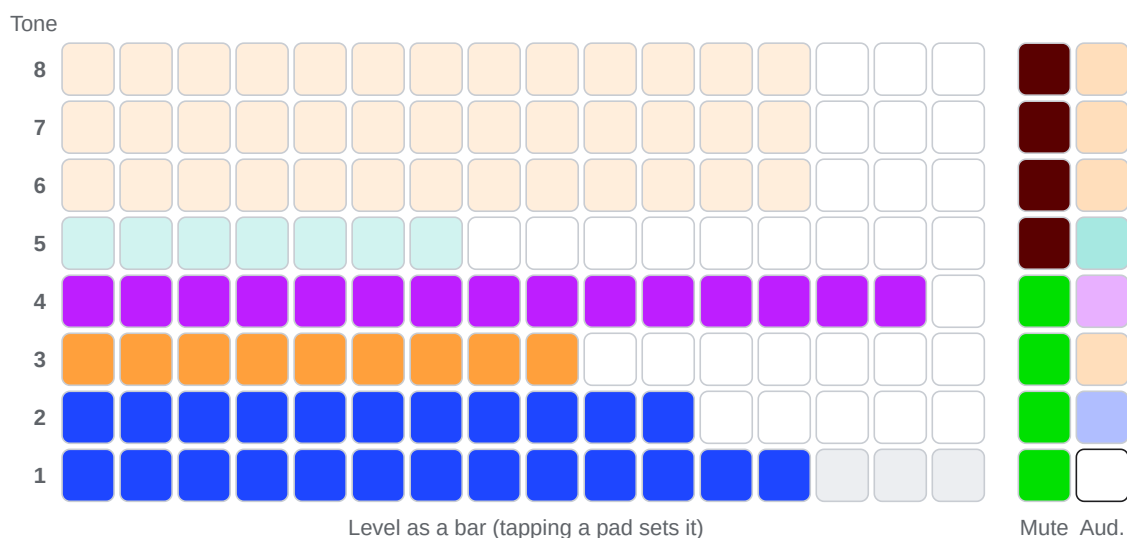
9.1 Quick start

1. In **song view**, press the **SCALE** button. The drone view opens and the Scale LED lights up.
2. In the bottom row, press the **mute pad** (first column to the right of the grid). It turns green and tone 1 sounds: a sine at 200 Hz.
3. Turn the **upper gold knob** to change the pitch: 1 Hz per detent, 10 Hz when turned fast.
4. Press the **KIT** button: the tone becomes *binaural*. On headphones you now hear a beating of 10 Hz.
5. Turn the **lower gold knob** to change the beat, or **press** it so the beat follows the tempo (1/16 at first).
6. Go back to song view with **BACK**, **SONG** or **SCALE**. The drone keeps sounding.

Alternatively, open the drone view in song view by pressing **SELECT** (song menu) → *Drone*. The arranger does not have this menu item, because the way back always leads to song view.

9.2 The drone view

The view is built like song view: one row per tone instead of one per clip, **tone 1 at the bottom**. Eight tones are visible at once. Use the **Y encoder** (vertical) to scroll to tones 9–16.



The pads of the drone view. Each row is a tone. The bar shows the level in the mode's color, followed on the right by the mute column and the audition column.

Pads in a row

- **Bar** = level of the tone, 16 pads for 0–50.
- **Tapping a pad** sets the level and selects the tone: pad 1 = 3, pad 8 = 25, pad 16 = 50.
- **Selected tone**: its row glows faint gray behind the bar.
- **Tones that are off** are strongly dimmed; their settings are kept.
- **Steady display**: the pads light evenly and do not flash with the beat.

Colors and side columns

- **Orange**: Tone (steady)
- **Blue**: Binaural
- **Turquoise**: Monaural
- **Violet**: Isochronic
- **Mute column**: ■ green = on, ■ dark red = off. Tapping toggles the tone and selects it.
- **Audition column**: tapping selects the tone. White = selected, otherwise the mode color, dimmed.

9.3 All controls

Control	Effect
Pad in the grid	set the level of this row's tone and select the tone
Mute pad	tone on/off, select tone
Audition pad	select tone
Turn Y encoder	scroll: tones 1–8 to 9–16
Turn X encoder	level of the selected tone in steps of 1 (0–50)
SHIFT + X encoder	pan of the selected tone (–32 left to +32 right)
Turn Select	pitch of the selected tone in tactile detents: in Hz 1 Hz per click, with SHIFT 0.1 Hz; as a note one semitone, with SHIFT one cent
SYNTH / KIT / MIDI / CV	mode of the selected tone: Tone / Binaural / Monaural / Isochronic. The button's LED shows the current mode.
Press Select	open the menu of the selected tone and of the whole drone (section 9.11)
Gold knobs in the Cutoff/Resonance section (pitch/beat, selected on opening)	
Turn upper gold knob, set in Hz	frequency: 1 Hz per detent; 10 Hz when turned fast; with SHIFT 0.01 Hz
Turn upper gold knob, set as a note	note in semitones; with SHIFT cents (±50)
Press upper gold knob	switch between Hz and note. The pitch stays (the nearest note plus cents, or the note's frequency).
Turn lower gold knob, beat free	beat: 0.1 Hz per detent, with SHIFT 0.01 Hz (0–100 Hz)
Turn lower gold knob, beat synced	note value: 1/1, 1/1T, 1/2, 1/2T ... 1/64, 1/64T (T = triplet)
Press lower gold knob	tempo sync on/off. Switching it on brings back the last note value used (1/16 at first). Switching it off brings back the beat in Hz.
Gold knobs in the other sections (press the mod button next to the knobs)	
Level/Pan	upper: level of the selected tone (0–50) · lower: pan (–32 to +32)
Attack/Release	upper: attack · lower: release of the selected tone's isochronic pulse (0–50, shown as 0–100 %)
Sidechain/Reverb	upper: sidechain ducking of the whole drone (0–50) · lower: reverb send of the whole drone (0–50)
Delay (v15)	upper: Life of the whole drone (0–50) · lower: Rate, how fast everything moves (0–50, 25 = default)
ModFX (v15)	upper: FM of the whole drone (0–50), press : FM form Sine or Saw · lower: Pulse width (5–50 %)
other mod buttons	no function; the section stays
BACK , SONG , SCALE	back to song view
PLAY , RECORD , tempo, save, load	as usual

The LED of the selected mod button lights up, and the knobs' LED rings show the values (for pitch/beat they stay dark). Pressing the gold knobs only works in the Cutoff/Resonance section and (upper knob) in the ModFX section. The selected section stays, even after the menu. In the drone view, the gold knobs act only on the drone, not on the song parameters. There is no undo in the drone view; drone settings are not part of the undo history.

9.4 Display: OLED and 7-segment



The OLED in the drone view: tone 1 is binaural at 100 Hz with a 4 Hz beat, which lies in the theta range. From the real v18.3 firmware in the emulator.

OLED

- **Title:** number and mode of the selected tone, e.g. 3: Binaural. If the tone is off, - off follows.
- **Large:** the pitch, 200.00 Hz or as a note, e.g. A2 +12c.
- **Bottom line:**
 - Tone mode: timbre and level, e.g. Sine Level 40
 - other modes: beat with frequency band, e.g. Beat 10.00 Hz Alpha
 - with tempo sync: 1/16: 8.00 Hz Alpha, as a triplet 1/16T: 12.00 Hz Beta
- Short pop-ups show level, pan, mode, Hz/Note, SYNC/Beat, the selected knob section and its values, e.g. Attack 12%.

7-segment

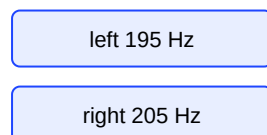
- **Frequency** with as many decimal places as fit: 55.25, 440.0, 1200.
- **Note** as a name, e.g. A2 (sharp shown as a dot).
- **While turning**, shown briefly:
 - Level: number
 - Pan: number
 - Beat: in tenths, e.g. 7.8
 - Note value: 16 for 1/16, 16T for the triplet

Note names follow the Deluge numbering: A3 = 440 Hz, middle C = C3.

9.5 The four modes

Binaural

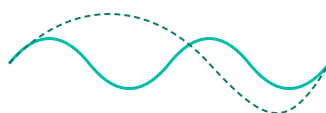
f = 200 Hz, Beat = 10 Hz



beating arises in the head

Monaural

both tones in both ears



beating audible in the sound

Isochronic

tone pulses with the beat



on/off with attack and release

Mode	What you hear	Remark
Tone	A steady sustained tone, the same left and right (mind the pan).	Beat and sync play no role.
Binaural	Left the frequency minus half the beat, right plus half the beat. The beating only arises in the head.	Works only with headphones ; over speakers it sounds like Monaural.
Monaural	The same two tones, but both in both ears. The beating is audible in the sound itself, also over speakers.	Good for rooms and recordings.
Isochronic	One tone that switches on and off with the beat. Attack and release set how softly it starts and stops.	Clearest effect, also over speakers.

Attack and release of the isochronic pulse

The pulse is on for half of each beat cycle. **Attack** is the time in which it rises at the start, **Release** the time in which it then fades. Both go from 0 to 50. Each step is 2 % of half the cycle length, so 50 is the whole half that is on. This way the pulse keeps its shape when the beat or tempo changes. An edge is never shorter than 1 ms, so nothing clicks. Both start at 6: at 10 Hz that is 6 ms, as in v12.

Attack / Release	Sound
0–6	clear, hard pulse
15–25	soft, without edges
50 / 50	almost sinusoidal swelling and fading
50 / 0	swells and cuts off, like a tone played backwards

A new mode or a new timbre is faded briefly through silence, so nothing clicks. Level, pitch, beat and pan glide smoothly to the new value.

Band	Delta	Theta	Alpha	Beta	Gamma
Beat	below 4 Hz	4 to below 8 Hz	8 to below 13 Hz	13 to below 30 Hz	30 Hz and above

The OLED shows the band for the current beat. The beat can be at most as high as the frequency itself, so the lower tone does not fall below 0 Hz.

9.6 Pitch: hertz or note

In hertz

10.00 to 5000.00 Hz, accurate to 0.01 Hz. Useful for fixed frequencies, for example 432 Hz, 528 Hz or a room resonance. A frequency in hertz stays fixed, even if you change the Master Tune.

As a note

Note 0–127 plus –50 to +50 cents. The tone **follows the Master Tune** (Settings → Tuning) like all other sounds of the Deluge. If you tune the Master Tune to an external instrument, the drone matches automatically.

If you switch between Hz and note with a knob press or in the menu (*Pitch as*), the pitch is kept. A frequency becomes the nearest note with cents; a note becomes its frequency at the current Master Tune.

9.7 Beat and tempo sync

The beat is the beating or pulse frequency of the Binaural, Monaural and Isochronic modes. Free, it runs from 0 to 100 Hz in 0.01 Hz steps; it starts at 10 Hz. With tempo sync it follows the song tempo, straight or as a triplet (three in the time of two, so 1.5 times as fast):

Note value	1/1	1/2	1/4	1/8	1/16	1/32	1/64
Beat at 120 BPM	0.5 Hz	1 Hz	2 Hz	4 Hz	8 Hz	16 Hz	32 Hz
Band	Delta	Delta	Delta	Theta	Alpha	Beta	Gamma
Triplet at 120 BPM	0.75 Hz	1.5 Hz	3 Hz	6 Hz	12 Hz	24 Hz	48 Hz

- **When the song is playing**, the pulse locks onto the bar grid (accurate to 1.2 ms in the test). This also applies with external MIDI clock and with sync scaling.
- **When the song is stopped**, the beat runs on freely at the set tempo.
- The beat follows **tempo changes** smoothly.

9.8 Timbres

Timbre	Sound
Sine	pure sine, clearest for beats
Soft	soft harmonics 2–5, warm
Organ	octaves like organ drawbars
Rich	harmonics 2–8, sawtooth-like and bright
Pulse (v15)	square wave with adjustable width (<i>Pulse width</i> 5–50 %, default 30): hollow and nasal when narrower, full at 50 %. With Life the width drifts slowly.

All timbres are band-limited: harmonics above 18 kHz are dropped, so nothing aliases or chirps on high tones. You choose the timbre in the menu (*Timbre*). Pulse applies to the single tone, the width to the whole drone.

9.8a Life and FM (v15)

A sustained tone soon sounds rigid. With **Life**, each tone moves by itself, as if played by hand; with **FM** it gets a color that blooms and fades again. Both apply to the whole drone, but each tone moves on its own paths.

Setting	Effect
Life 0–50	Each tone drifts on slow random paths that never repeat. At 50: pitch ± 6 cents (path at about 0.08 Hz), volume ± 3 dB ("breath"), brightness and stereo position ("shimmer"). Everything grows evenly with Life. On a beating tone, both sides drift together: the beat stays exactly as set.
Rate 0–50	How fast Life and the blooming of the FM move. 25 as designed; every 12.5 steps doubles or halves the speed.
FM 0–50	A modulator at the tone's frequency plus 0.3 Hz colors it. The depth blooms and fades, each tone on its own path. On high tones it is limited, so nothing audibly aliases.
FM form	Sine: soft, bell-like · Saw: brighter, more tinny. Switching crossfades.

- **At Life 0 and FM 0** (and without Pulse) the drone sounds exactly as in v13 and v14.
- Changes to Life, Rate and FM glide; nothing clicks. A switch to or from Pulse goes through silence, like any change of timbre.
- Life only moves what the drone itself does. Tempo sync, sidechain and Master Tune still apply as set.

9.9 Level, pan, volume and mix

- **Level** per tone, 0–50 in 1 dB steps, 0 is silent. A tone at level 50 with drone volume 50 sits at -12 dBFS. New tones start at level 40.
- **Pan** per tone, -32 to $+32$, equal power: 0 dB in the center, up to $+3$ dB on one side at the far ends. With Binaural, pan acts as a balance between the ears: panned hard, one ear barely hears its tone and the beating is lost.
- **Volume** (in the menu): volume of the whole drone, 0–50 in 1 dB steps, 50 = 0 dB, starts at 40.
- **Song volume**: the drone follows it, like the metronome.
- **In the mix** the drone is added after the song effects, so without delay and song filter. It can be heard over USB audio (v8).
- **Reverb** (menu *Reverb amount* or lower gold knob in the Sidechain/Reverb section): send of the whole drone into the song's reverb, 0–50. At 50 it sends as much as a sound with the reverb knob turned fully up. The send follows level, pan and sidechain ducking and glides smoothly when turned.

Many loud tones together can clip: 16 tones at level 50 add up to far more than full scale. For pad sounds of many tones, lower the level per tone or the drone volume.

9.10 Sidechain

The drone can duck under the song's sidechain triggers, for example under a kick with *Send to sidechain*. The level glides from sample to sample, so nothing clicks.

Menu item (Sidechain)	Effect
Volume ducking	Amount 0–50, off at 0. At the lowest point of a full hit, the level drops linearly: 10 → -2 dB, 20 → -4 dB, 30 → -8 dB, 40 → -14 dB, 50 → fully gone. While turning, the OLED shows the depth, e.g. Sidechain 30 -8dB.
SYNC	envelope in tempo, as with the other sidechains
ATTACK / RELEASE	how fast the drone ducks and comes back
SHAPE (title <i>Sidech. shape</i>)	shape of the curve, as with the reverb; only visible when Volume ducking is above 0

9.11 The menu

Pressing **SELECT** opens the *Drone tone* menu for the selected tone. Items that do not apply at the moment are hidden. The names are written as the Deluge shows them.

Item	Values	Visible
MODE	Tone, Binaural, Monaural, Isochronic	always
Pitch as	Hz, Note (switches at the same pitch)	always
Frequency	10.00–5000.00 Hz	with Hz
Note	0–127, shown as a note name	with Note
Cents	–50 to +50	with Note
Beat	0.00–100.00 Hz	not with Tone, and sync off
SYNC	Off, 1/1, 1/1T to 1/64, 1/64T	not with Tone
ATTACK	0–50 (shown as 0–100 %)	with Isochronic
RELEASE	0–50 (shown as 0–100 %)	with Isochronic
Timbre	Sine, Soft, Organ, Rich, Pulse	always
Level	0–50	always
PAN	–32 to +32	always
Volume	0–50, whole drone	always
Reverb amount	0–50, send of the whole drone	always
Life	0–50, whole drone (section 9.8a)	always
Rate	0–50, whole drone, 25 = default	always
FM	0–50, whole drone	always
FM form	Sine, Saw	always
Pulse width	5–50 %, whole drone	always
Sidechain	submenu (section 9.10), whole drone	always

BACK leaves the menu and takes you back to the drone view.

9.12 Saving, loading, export

- **Saving:** the drone is saved with the song (tag <drone>), but only if it has been set up. Older firmware simply skips it.
- **Loading:** when you load another song or use *Clear Song*, the drone fades out in about 60 ms. The new song's drone starts from silence.
- **Stem export:** the drone is not in the stems. Afterwards it starts again from silence.
- **Recording:** you record it with the rest of the song via USB audio or the output.

9.13 With external devices

Set the tones to **Note** and tune the **Master Tune** (Settings → Tuning) to your device, for example a volca keys. Then the drone and the instrument share the same tuning. If you move the Master Tune later, all tones set as notes follow; tones in Hz stay fixed.

9.14 Recipes

Goal	How to
Calm alpha bed	Tone 1: Binaural, Sine, 200 Hz, Beat 10 Hz, Level 35. Tone 2: Binaural, Soft, 300 Hz, Beat 10 Hz, Level 28. Listen on headphones.
Pulse in the groove	Isochronic, Sync 1/16 (8 Hz at 120 BPM), Rich, one octave below the song's root note. Add sidechain with Volume ducking 20.
Soft triplet pulse	Isochronic, Sync 1/8T, Soft, Attack 20, Release 30. Shape it live with the gold knobs in the Attack/Release section.
Drone in the room	Monaural, Organ, Reverb amount 20–30: the drone sits in the same room as the song.
Root-note pad	Three tones as notes: root, fifth, octave; Timbre Organ, mode Tone, Level 30–35. The tones follow the Master Tune.
Floating choir	Monaural, Soft, the same note on two tones, Beat 0.3 and 0.5 Hz, Pan –20 and +20.
Slow movement	Binaural or Monaural with Sync 1/1 or 1/2: one beating per bar or half bar.
Breathing pad (v15)	Three Soft tones as root, fifth, octave; Life 25, Rate 20. The pad moves slowly by itself, each tone a little differently.
Bells in the fog (v15)	Monaural, Sine, two high tones; FM 30 with form Sine, Life 15, Reverb amount 25.
Rough bass drone (v15)	Tone, Pulse, one low note; Pulse width 15, FM 20 with form Saw, Life 10. Put it under the kick with sidechain.

9.15 Specifications and limits

Property	Value
Tones	16, each with mode, timbre, pitch, beat, sync, level, pan
Frequency / note	10.00–5000.00 Hz · note 0–127 ± 50 cents
Beat	0–100 Hz free (accurate to 0.01 Hz) or 1/1–1/64 synced, also as triplets
Isochronic pulse	Attack and release 0–50 (each step 2 % of half the cycle length), at least 1 ms
Reverb send	0–50 for the whole drone
Level, Volume, Pan	0–50 (1 dB), 0–50 (1 dB), –32 to +32
Purity	Sine: spurious products below –108 dB
Life, FM, Pulse (v15)	Life and FM 0–50 and Rate 0–50 for the whole drone, FM form Sine or Saw, pulse width 5–50 %
CPU time (Cortex-A9, counted in the emulator)	1 sine tone 0.35 %, 1 binaural tone 0.5 % (0.6 % from v15 on), 16 binaural tones with harmonics 6.9 % of the CPU. With v15 per binaural tone: Life 0.7–1.0 %, FM 1.3 %, Pulse 1.0 %, all together 2.1 %; 16 tones with everything just over 30 %. The drone is never culled like synth voices.
Memory	Wavetables 27 KB internal RAM, with v15 another 14 KB (sawtooth tables for Pulse and FM)

- **Tested** in the emulator (46 checks on PC and Cortex-A9) and on the device with v12. The new features of v13 and v15 are not yet tested on the device; Life, FM and Pulse are checked on the PC with 116 checks.

- The song drone sends no MIDI and takes no notes from the keyboard or MIDI. Drone tracks (section 9.16) take notes: they open the row's gate.
- The song drone's values cannot be automated. With drone tracks, the pitch of each row is recorded with REC as a Hz track (section 9.16).

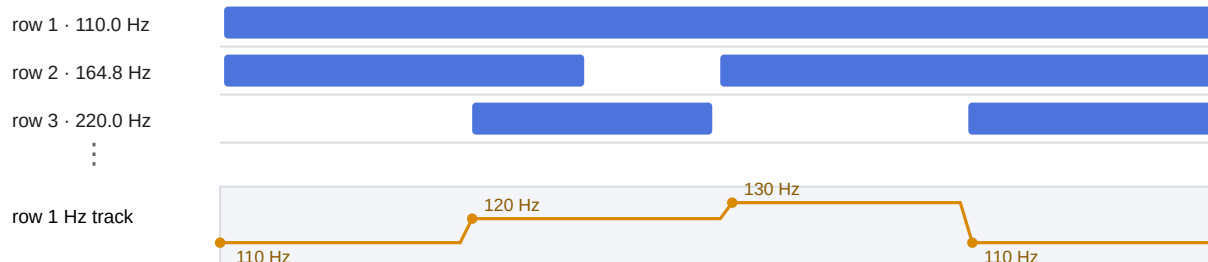
9.16 Drone tracks (v16)

The song drone sounds as long as it is on. With **drone tracks** you place drones where you want them: a drone track is a kit whose rows are drone tones. You launch their clips in song view and place them in the arranger like any kit clip. Several drone tracks can overlap, each with its kit's effects, volume and sidechain.

What	How to
Make a drone track from the drone	In the drone view, SHIFT + KIT . This creates a new kit (DRONE1, DRONE2 ...) with 16 rows, the tones of the song drone; the tones that are on get a note across the whole clip. The clip sits at the bottom of song view and is not launched yet. The song drone itself stays unchanged.
Turn a kit row into a drone row	In a kit's clip view, hold the row's audition pad (or create a new row), then SHIFT + KIT . The row sounds at 200 Hz.
Gate	A drone row's notes are its gate: as long as a note lasts, the tone sounds, with a soft fade in and out. A note across the whole clip holds over the loop boundary. With the row's arpeggiator, the tone pulses in rhythm.
Select a row	Its audition pad (Affect Entire off). If the row is sounding from the sequencer at the time, the pad only selects it; it keeps sounding.
Pitch	Turn Select: 1 Hz per detent, with SHIFT 0.1 Hz. Upper gold knob: 1 Hz, 10 Hz when turned fast, with SHIFT 0.01 Hz. A tone set as a note: semitones, with SHIFT cents.
Level	Lower gold knob
Row menu	Press Select: mode, pitch, beat, sync, pulse, timbre, level, pan, plus the row's arpeggiator and randomizer
Display	OLED: mode and Hz as the row's name; 7-segment: the Hz

Kit DRONE1, clip view

one loop →



A drone track: a kit whose rows are drone tones. Notes switch the tones on and off (blue). Turning the pitch while **RECORD** is on writes nodes into the row's Hz track (orange); between nodes the tone glides in about 15 ms. The Hz values are an example.

Recording and playing Hz changes

- **PLAY**, then **RECORD**, then turn Select or the upper gold knob: each change is written as a node into the row's **Hz track** and plays back in the next loop. Between two nodes the tone glides in about 15 ms. This also works when recording into the arranger.
- Without **RECORD**, the row's base pitch changes; a recorded Hz track shifts with it.

- While **RECORD** is on: **SHIFT** + Select records the fine 0.1 Hz steps. **SHIFT** + upper gold knob deletes the Hz track, like any automation; the row goes back to its base pitch at once.
- The Hz track belongs to the row, not to the notes: recording notes, moving them or spreading them in a Euclidean pattern leaves it unchanged.
- MIDI pitch bend on a drone row acts within the row's bend range (default 2 semitones) and is also recorded while **RECORD** is on.

Good to know

- CPU time: about 0.3–0.45 % CPU per sounding row, 16 rows 5–7 %. Rows with a closed gate cost nothing.
- A new drone kit sounds as loud as the song drone at the same setting.
- Older firmware opens songs with drone tracks without the drone rows; the other rows stay correct. If you save them there, the drone rows are gone.

10. The drone with a Korg volca keys

The volca keys is a small analog synthesizer with three oscillators, one filter and one envelope. Played from the Deluge, it delivers long, warm tones that move slowly. The Deluge's **Frequency Drone** lays exact tones and beatings underneath: binaural, monaural or isochronic. This manual shows you how to wire, tune, play and record both.

This chapter applies to mastertune v16 and later, including v18.3. The volca keys is fixed at A = 440 Hz; section 10.5 shows how it fits with the master tune.

Start quiet. Sustained tones, especially low ones, can seem louder than expected on headphones. Start with a low volume on the volca and the Deluge and raise it slowly.

Sources: The information on the volca keys is based on Korg's MIDI chart (v1.02), excerpts from the Korg manual, reviews and forums. Points marked **verify** are uncertain or have conflicting sources. Check them on the device or in the Korg manual before you rely on them.

10.1 What the combination is good for

What the volca keys adds

- Analog sound that drifts slightly and so feels alive
- Three oscillators: as a chord of up to three notes (Poly) or as one wide tone (Unison, Octave, Fifth)
- Filter, LFO and delay for slow movement
- Remote control via MIDI: notes, pitch bend and 16 control changes (CC)

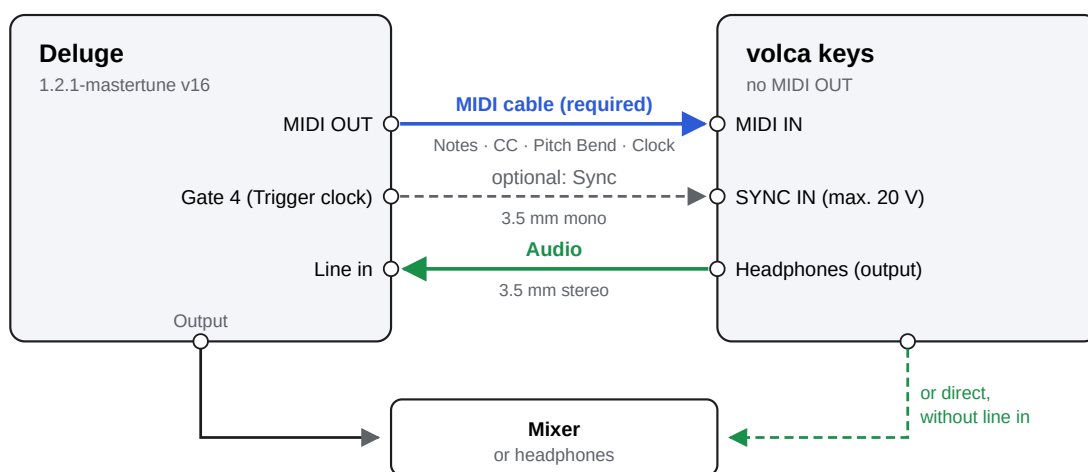
What the Deluge adds

- Long notes in a MIDI clip, CC automation, MIDI clock
- The Frequency Drone with exact tones: Tone, Binaural, Monaural, Isochronic, with sidechain
- Drone tracks (v16) that you launch like clips
- Recording the volca via the line in, also as an audio clip that follows the Master Tune

Typical uses: an analog pad over a binaural foundation, a filter sweep over many bars, a slow beating between the drifting analog tone and the exact drone tone, or a recording of the volca that you then retune to 432 Hz.

Rule of thumb: The volca keys is fixed at A = 440 Hz. Leave the Master Tune at **440.0** and set the drone tones to **Note**. Then the volca and the drone match. For 432 Hz and other tunings, see section 10.5.

10.2 Wiring



Basic setup. Only the MIDI cable is required. Sync and line in are optional. Without line in, the volca goes straight to the mixer.

Connection	Cable	Purpose
Deluge MIDI OUT → volca MIDI IN	MIDI cable. That the volca has a 5-pin DIN socket is very likely verify	Notes, pitch bend, CCs and MIDI clock. This is the only connection you need.
Deluge Gate 4 → volca SYNC IN	3.5 mm mono	Only if the volca's sequencer should run from pulses. For this, Gate 4 must be set to <i>Clock</i> (section 10.4). Not needed for notes from the Deluge. SYNC IN takes at most 20 V.
volca headphones → Deluge line in	3.5 mm stereo, plug to fit the line-in socket	Recording as an audio clip, input as an oscillator, resampling (section 10.8)
or volca headphones → mixer	3.5 mm stereo	if the volca should not run through the Deluge

- **No return path:** The volca keys has no MIDI OUT (only as a mod). So the Deluge cannot record knob movements on the volca as MIDI. Automation is always created on the Deluge.
- The volca's **SYNC OUT** (3.5 mm mono, 5 V, 15 ms pulse) is not needed in this setup.
- **Power:** 6 × AA last about 10 hours. For long drone sessions a 9 V power adapter (DC) is safer.

10.3 Setting up the volca keys

According to the Korg manual, you reach the volca's global settings by **switching it on while holding the FUNC button**. The parameters are then on the step buttons: **1** Auto Power Off, **2** battery type, **3** SYNC OUT polarity, **4** SYNC IN polarity, **6** MIDI clock, **7** MIDI ShortMessage. **REC** saves and restarts, **PLAY** cancels without saving.

Setting	For this setup	Remark
MIDI channel	leave at 1 , the default	The Deluge clip then also goes to channel 1. The sources differ on exactly how to change the channel: buttons 1–16 directly, via MEMORY or with FUNC + button 16 verify
MIDI clock	LED of step 6 on = Auto (follows MIDI clock), off = internal	Only matters if the volca's sequencer or Motion Sequence should run at the Deluge's tempo. Under MIDI clock, the volca's tempo knob has no effect.
SYNC IN polarity	rising edge (default)	global parameter on button 4; only matters with a sync cable
Auto Power Off	off for long drones	Button 1. Otherwise the volca switches itself off after about 4 hours.
MIDI ShortMessage	leave at default	Button 7. What exactly the parameter does is not documented verify . If the volca does not respond to notes or CCs from the Deluge, switch it.

If a cable is plugged into SYNC IN, the volca's sequencer follows only the pulses and ignores its own clock. The Korg manual also lists the tempo range as a global parameter. It plays no role in this setup, because the tempo comes from the Deluge.

10.4 Setting up the Deluge

MIDI clip for the volca

Setting up a MIDI clip is part of the standard operation of 1.2.1. The mastertune documents do not describe it; the official Deluge manual explains it in detail. In short:

1. Create a new clip as usual and switch to clip view.
2. Press the **MIDI** button. The clip now plays to a MIDI channel, and the MIDI LED lights up.

3. Turn Select until the volca's channel appears (default: 1).

4. Enter notes. For drones: long notes, see section 10.6.

Clock and tuning

Menu	What you set
Settings → MIDI → Clock → Output Title "MIDI clock out"	On if the volca should run along at the tempo (its MIDI clock set to Auto). Since v15 the MIDI clock output also keeps running when the Deluge itself follows an external MIDI clock.
Settings → Gate → Gate output 4 → Clock Title "Gate out4 mode" Settings → Trigger clock → Output → PPQN Title "Output PPQN"	Only with a sync cable. The trigger clock comes from Gate 4 , but only if its mode is set to Clock . The default is V-trig, and then the sync cable stays silent. Set <i>Output PPQN</i> to 2 : the sources point to 1 pulse per 2 steps verify . Compare the volca's tempo with the Deluge's. MIDI clock without a sync cable is simpler.
Settings → Tuning → Master tune (Hz) 7-segment: TUNE → MTUN	Leave it at 440.0 for the volca (section 10.5). Range 415.3 to 466.2 Hz in 0.1 Hz steps.

10.5 Tuning: volca and Master Tune

The volca keys has **neither a Master Tune nor a fine tuning**. So A = 440 Hz is effectively fixed. The Deluge's Master Tune therefore shifts everything in the Deluge, but not the volca.

Sound source	Follows the Master Tune?
Drone tones as Note	yes
Drone tones in Hz	no, the frequency stays fixed
Deluge synths, kits, samples	yes
Audio clips, including recordings of the volca	yes, by time-stretch; the tempo stays in sync. A recording is only shifted by the difference from the tuning it was recorded at, never twice.
volca, played via MIDI	With a tuning other than 440 Hz, the Deluge sends RPN 1 (Channel Fine Tuning) on its channel. That the volca evaluates RPN 1 is not documented and, given the missing fine tuning, unlikely verify . Expect it to stay at 440.
volca via line in as an oscillator	no, it sounds unchanged in its own tuning

The volca's auto-tuning

The volca retunes itself: after about **10 seconds of silence** it corrects the drift of its oscillators. This runs continuously; no warm-up time is needed. For drones this means [verify](#): as long as a sustained tone sounds, it does not retune and can drift slightly over time. Give it more than 10 seconds of silence between two pieces, for example by stopping with **PLAY** on the Deluge.

At 440 Hz (recommended)

Master Tune at 440.0, and all drone tones that should match the volca set to **Note** (press the upper gold knob or use *Pitch* as in the menu). This is the approach from section 9.13: tune the Master Tune to the external device. For the volca that means 440.0. If the Master Tune has never been set to anything else since power-on, the Deluge also sends no RPN 1.

At 432 Hz: honest options

From 440 to 432 Hz is −31.77 cents. The volca cannot do this by itself. There are three ways:

Way	How to	Drawback
A. Record, then retune (safest way)	Master Tune at 440.0 , record the volca as an audio clip (section 10.8). Then Master Tune to 432.0: the clip sounds 31.77 cents lower, the tempo stays. Drone tones set as notes follow.	Not live. Time-stretch needs more CPU time and has very quiet artifacts. Important: A recording made at Master Tune 432.0 counts as 432 material and would stay at the volca's pitch, that is 440.
B. Pitch bend	Lower the volca by about 32 cents with pitch bend. With a bend range of ± 2 semitones, that is about 16 % of the way down (14-bit value about 6890 instead of 8192). Check with a tuner.	The bend range comes from a forum. Resolution and interaction with the auto-tuning are not documented verify . How the Deluge holds a fixed bend value in a MIDI clip is not in the mastertune documents (official manual). Any further bend adds on top.
C. Deliberately against each other	Master Tune 432.0, the volca stays at 440. The same notes then beat audibly against each other: at A2 by 4 Hz, at A3 by 8 Hz.	Only useful as an effect, not as a tuning.

A drone tone at **432 Hz in Hz** is not the same as Master Tune 432.0: it stays fixed, even if you change the Master Tune. At Master Tune 432.0 the note A3 gives exactly 432 Hz.

10.6 Playing drones on the volca

Long notes

- Enter long notes in the MIDI clip, for a pad across the whole clip. The volca holds the tone as long as the note lasts.
- **A new note restarts the envelope for all held notes** (secondary source). So for a calm pad, start all notes of a chord at the same time. Staggered entries create a new attack each time.
- A note that starts at step 1 and fills the whole clip is held by the Deluge across the loop boundary. It sends neither Note Off nor Note On there, so the volca does not retrigger.
- In Poly mode the volca plays up to three notes. What happens with more than three notes is not documented [verify](#). Stick to three at most.

Voice modes (VOICE)

Mode	Sound	For drones
Poly	3 notes, sawtooth	chord of up to three notes, for example root, fifth, octave
Unison	mono, all three oscillators on one note	wide, carrying single tone; beating with Detune
Octave	mono	single tone with an octave sound
Fifth	mono	single tone with a fifth sound, open and empty
Unison Ring	square wave with ring modulation	metallic, bell-like tone
Poly Ring	square wave with ring modulation	metallic chord

Via MIDI, CC 40 selects the mode. Which values give which mode is not documented [verify](#). So set the mode with the volca's knob. The same applies to the octave setting (CC 41, 6 octaves on the knob).

Envelope for sustained tones

- **Attack** high: the tone fades in slowly.
- **Sustain** fully up: the tone holds its volume. That it then holds indefinitely is inferred [verify](#).
- **Decay/Release** long: the tone fades out softly.
- There is only **one** envelope and **one** filter (12 dB) for all three voices.

Detune and portamento

- **Detune** (CC 42) detunes the oscillators against each other. In Unison this creates an inner beating that makes the tone wide.
- **Portamento** (CC 5) makes the tone glide from note to note. With long notes this becomes a slow drift.

Slow filter sweeps

- **Via CC:** automate CC 44 (Cutoff) in the MIDI clip over many bars. How to record or draw CCs in a MIDI clip is part of the standard operation of 1.2.1 (official manual). The volca has no MIDI OUT, so the route always goes through the Deluge.
- **Via LFO:** LFO Rate (CC 46) low, LFO Cutoff Int (CC 48) to taste. Waveforms sawtooth, triangle, square. The LFO can be synced to the start of the note (Trigger Sync).
- **Steps:** according to a review, the filter steps audibly at high resonance. For very slow sweeps, keep the resonance moderate.
- **Delay:** Delay Time (CC 52) and Feedback (CC 53) add space. Feedback also sets the amount of delay in the sound.

10.7 Together with the Frequency Drone

You open the song drone in song view with [SCALE](#) or via [SELECT](#) → song menu → *Drone*. Chapter 9 describes how to operate it. This chapter is about how the two play together.

Hitting the same notes

- Set the drone tone to **Note**: press the upper gold knob or use *Pitch* as in the menu. Turning the upper gold knob selects the note, with [SHIFT](#) the cents.
- The note names follow the Deluge numbering: A3 = 440 Hz, middle C = C3. Because the Deluge sends the volca its notes, this numbering applies equally to the MIDI clip and the drone.
- For a foundation, place the drone one or two octaves below the volca's root note. For warmth, a fifth or octave above works well.

Binaural beat under an analog chord

The volca plays a chord in Poly mode. Below it are one or two drone tones in **Binaural** mode ([KIT](#) button) on the root note, one or two octaves lower. The beating arises only in the drone; the volca provides the pad above it. This works **only with headphones**.

Beating between analog and digital

If you put a drone tone on the same note as the volca and detune it by a few cents, the two tones slowly beat against each other. The beating is about $frequency \times cents \div 1731$: at A2 (220 Hz) and 3 cents, about 0.4 Hz. Because the volca drifts slightly, the beating drifts with it. This sounds alive.

Sidechain

The drone can duck under the song's sidechain triggers, for example under a kick with *Send to sidechain*. In the drone menu under *Sidechain* you find *Volume ducking*, SYNC, ATTACK/RELEASE and SHAPE. The volca does not duck along; its pad stays steady while the drone pumps underneath.

Drone tracks (v16)

In the drone view, [SHIFT](#) + [KIT](#) creates a kit DRONE1 (DRONE2 ...) from the drone tones. The rows' notes are the gate; a row's arpeggiator makes the tone pulse. You launch the clip in song view together with the volca's MIDI clip. That way the volca and the drone start and end together.

Level: a drone tone at Level 50 and Volume 50 sits at -12 dBFS. The volca comes in via the line in or the mixer. Balance the two by ear and start with a drone Level of about 30.

10.8 The volca through the Deluge

With a cable from the volca's headphone output into the Deluge's line in, you can record the volca or send it through the Deluge. Start with a low volume on the volca. The official Deluge manual describes in detail how to use audio clips and recording. The menu items are called *Audio source*, *Input (left/right/stereo)* and *Monitoring*.

Route	What happens	Tuning
Audio clip	An audio track with <i>Audio source</i> set to the input records the volca. With <i>Monitoring</i> you hear the input while recording.	The recording then follows the Master Tune. If the tuning is not 440 while recording, the WAV file gets an mtun block and is never tuned twice.
Input as oscillator	A Deluge synth uses the line in as an oscillator. This way the volca runs through the Deluge's filters, envelopes and effects.	does not follow the Master Tune, because the volca itself is already tuned
Resampling	The Deluge records its own output, with the volca via the line in.	like an audio clip, with an mtun block if Master Tune $\neq$ 440

- **Drone in recordings:** the song drone can be heard via the output and USB audio and also ends up in **resampling recordings**. It is **missing from stems**.
- **432 Hz:** for way A from section 10.5, record at Master Tune 440.0 and only change it afterwards.

10.9 Recipes

Values on the volca are knob positions by ear. The Deluge values come from chapter 9. All recipes assume MIDI channel 1 and Master Tune 440.0, except recipe 5.

1. Analog chord over a binaural base

1. volca: VOICE to **Poly**, Attack about half, Sustain fully up, Decay/Release long, Cutoff half.
2. Deluge: MIDI clip of 4 bars, with C3, E3 and G3 together across the whole clip.
3. **SCALE** in song view. Tone 1: mute pad on, **KIT** (Binaural), *Pitch* as Note, note C2, Timbre Sine, Beat 6 Hz, Level 32.
4. Tone 2: Binaural, note G2, Timbre Soft, Beat 6 Hz, Level 26.
5. **PLAY**. Listen on headphones.

2. Unison drone with a filter sweep over 16 bars

1. volca: VOICE to **Unison**, Detune about a quarter, Attack long, Sustain fully up, resonance moderate.
2. Deluge: MIDI clip of 16 bars, one note A1 across the whole clip.
3. Automate CC 44 (Cutoff) in the clip: from low to high over 8 bars, back over 8 bars.
4. Drone tone 1: mode Tone (**SYNTH**), *Pitch* as Note (press the upper gold knob), note A0, Timbre Sine, Level 30. It carries the low end while the filter moves.

3. Slow beating against the volca

1. volca: **Unison**, Detune fully down, Sustain fully up. MIDI clip with one long note A2.
2. Drone tone 1: mode Tone, Timbre Soft, *Pitch* as Note (press the upper gold knob), note A2, then turn **SHIFT** + upper gold knob to +3 cents.
3. Set Level so that the drone and the volca are about equally loud. This creates a beating of about 0.4 Hz.
4. Optional: Life 10 (Delay section, upper gold knob) for extra movement.

4. Pulsing ring drone in tempo

1. volca: VOICE to **Poly Ring**, Sustain fully up, Delay Time medium, Feedback low.
2. Deluge: MIDI clip with two long notes a fifth apart, for example D2 and A2.
3. Drone tone 1: **CV** (Isochronic), *Pitch* as Note (press the upper gold knob), note D1, Timbre Soft. Press the lower gold knob (SYNC) and turn it to 1/8T. Attack/Release section: Attack 20, Release 30.
4. Add a kick with *Send to sidechain*, and set *Volume ducking* to 20 in the drone menu.

5. Bringing the volca to 432 Hz

1. Settings → Tuning → Master tune (Hz) to **440.0**.
2. Set up the volca as in recipe 2 and record it via the line in into an audio clip, for example 8 bars.
3. Master Tune to **432.0**. The audio clip now sounds 31.77 cents lower; the tempo stays.
4. Drone tones set as notes follow; A3 is now at 432 Hz. Tones in Hz stay fixed.
5. Do not play the volca live along with it any more: it is still at 440.

10.10 CC table

Based on Korg's MIDI chart (v1.02). By the MIDI standard, CC values range from 0 to 127. According to Korg, some parameters can only be reached via MIDI (marked with * in the chart). The Korg chart itself shows which ones.

CC	Parameter	For drones
5	Portamento	gliding between long notes
11	Expression	check the effect on the device verify
40	Voice	voice mode; value ranges not documented, choose it on the knob verify
41	Octave	octave setting; value ranges not documented verify
42	Detune	width and inner beating, especially in Unison
43	VCO EG Int	pitch follows the envelope; use little for calm drones
44	Cutoff	the knob for slow filter sweeps
45	VCF EG Int	how far the envelope opens the filter
46	LFO Rate	low for slow movement
47	LFO Pitch Int	vibrato; a little makes the tone come alive
48	LFO Cutoff Int	the filter breathes with the LFO
49	Attack	fade in
50	Decay/Release	fade out
51	Sustain	fully up for sustained tones
52	Delay Time	space
53	Delay Feedback	repeats and amount of delay

Other MIDI messages	Status
Note On/Off	yes
Velocity	evaluated, according to a forum verify
Pitch Bend	yes; range ± 2 semitones according to a forum, not changeable verify
Program Change, RPN 1 (Fine Tuning), Local Control	not documented, probably not available verify

The list contains no CC for resonance. The Frequency Drone itself sends no MIDI, takes no notes and cannot be automated. All CCs to the volca come from the MIDI clip.

10.11 Troubleshooting

Problem	Cause and solution
The volca does not play	Cable in the Deluge's MIDI OUT and the volca's MIDI IN? Clip on the same channel as the volca (default 1)? Clip launched and not muted? The volca's volume turned up? Switch the global parameter MIDI ShortMessage (button 7) verify .
The volca's tempo knob does nothing	It follows the MIDI clock (LED of step 6 on) or a sync cable. This is intended.
The volca's sequencer runs twice or half as fast	With a sync cable: <i>Output PPQN</i> does not match. Try 4 instead of 2, or use MIDI clock without a cable. If it does not run at all: Gate 4 set to <i>Clock</i> ?
The pad sound restarts	A new note restarts the envelope for all notes. Start chord notes at the same time. At the loop boundary: the note must start at step 1 and fill the whole clip.
A note is missing from the chord	More than three notes at once, or a mono mode is active. VOICE to Poly, at most three notes.
The volca and the drone do not match	Master Tune not at 440.0? The volca does not follow it. Drone tones in Hz instead of as notes? After playing for a long time the volca can drift: give it more than 10 s of silence, then it retunes.
A recording sounds too high at 432 Hz	It was made at Master Tune 432.0 and so counts as 432 material. Record it again at 440.0.
A filter sweep sounds steppy	At high resonance the filter steps audibly. Lower the resonance or make the movement with the LFO.
The volca suddenly switches off	Auto Power Off after about 4 hours, or empty batteries. Turn off the parameter on button 1, use a power adapter.

10.12 Limits and open points

Limits

- The volca keys has no fine tuning. The Deluge's Master Tune does not retune it; 432 Hz only works via recording or pitch bend (section 10.5).
- Paraphonic: three voices share one filter and one envelope. There are no independent envelopes per note.
- No MIDI OUT: the volca can send neither notes nor knob movements to the Deluge.
- Auto-tuning only in pauses: with sustained tones lasting hours, the volca can drift slightly.

Open points [verify](#)

Not documented, or with conflicting sources: the MIDI IN socket as 5-pin DIN (very likely); the exact way to set the MIDI channel; the pulses per step at SYNC IN (probably 2 PPQN); the value ranges of CC 40 and CC 41, the effect

of CC 11 and which parameters can only be reached via MIDI; pitch bend range, velocity, Program Change, RPN 1 and Local Control; the behavior with more than three notes and indefinite holding with Sustain fully up; whether a sustained tone suppresses the auto-tuning. On the Deluge side: a fixed pitch bend value in a MIDI clip.

Sources on the volca: Korg's MIDI chart v1.02, product page korg.com (connections, power), excerpts from the Korg manual, Sound On Sound, Sweetwater, sonicstate, zZounds, korgforums.com. On the Deluge: README and drone manual of 1.2.1-mastertune, source code v16 (menu names, Gate 4 as trigger clock and its mode, long notes at the loop boundary, drone during resampling).

11. Reference

11.1 New and changed menu items

Menu item	Values	7-segment	Since	See
Settings				
Tuning → Master tune (Hz)	415.3–466.2 Hz in 0.1 Hz steps, default 440.0	TUNE → MTUN	v2	2.1
Community features → USB audio	off (default), on; acts after a restart	UAUD	v8	4.2
Community features → Kit RAM saver	on (default), off	KRAM	v9	3.3
Community features → Flicker-free dimming	on (default), off		v13	7.5
Community features → Output limiter	off (default), on	LIMT	v18	5.8
Community features → Filter crossing guard	off (default), on	CR0S	v18	5.5
CPU monitor	Off, On, Alerts, Profile	CPU	v13	3.4
OLED brightness	1–10, 10 = as before		v18	7.4
Firmware version	readable, without hash		v18	1.4
Reverb (song and sound)				
Model	adds Digital	DIGI	v10	5.10
Damping	0 bright ... 50 dark (Mutable now the right way round)		v10	5.10
HPF	0 = 20 Hz, 25 = 190 Hz, 50 = 540 Hz		v10	5.10
LPF	0 = 500 Hz ... 49 = 18.6 kHz, 50 = off (default)		v10	5.10
Modulation (after LPF)	0–50, default 0		v14	5.10
Pre-delay	0–100 ms, default 0		v14	5.10
Delay (sound, kit, audio track, song)				
Time change (after Type)	Fade (default), Tape		v14	5.11
LPF, HPF (after Sync)	LPF 50 = off (default); HPF 0 = off (default)		v11	5.11
Arpeggiator (synth, kit row, MIDI and gate rows)				
Preset	in the list, with Walk		v5	6.1
Latch	off, on		v5	6.2
Step repeat	1–8		v5	6.1
Note mode	adds Walk 1–3, Pattern		v5	6.1
Ratchet notes	Auto, 2–8, Roll		v6, v13	6.3
Ratchet bounce	–10 ... +10		v5	6.3
Bounce length	1–16		v13	6.3

Menu item	Values	7-segment	Since	See
Bounce velocity	Even, Fade, Rise		v6, v13	6.3
Randomizer (own menu after Arpeggiator)	Lock, spreads, probabilities		v5	6.5
Kit arpeggiator (kit, Affect Entire); Include in kit arp (row)			v5	6.1
Everywhere				
Every volume and level	0.5 dB per detent, shown in dB, OFF at the bottom	e.g. -3.5	v18	5.7
Song menu (SELECT in song view) → Drone	opens the drone view		v12	ch. 9

11.2 Buttons and shortcuts

Where	Press	What happens
any view, also playing	hold LEARN , press TEMPO	CPU monitor off, or back to the last mode (§3.4)
Master tune (Hz)	horizontal encoder	moves the cursor between the 1 Hz and the 0.1 Hz digit (§2.1)
song browser, group line	SELECT or LOAD	opens the group (§7.1)
song browser, a version	BACK	closes its group (§7.1)
song browser, a version	SHIFT + SAVE	deletes; blocked on a closed group line (§7.1)
song change armed	gold knobs, mod buttons	act on the master effects of the playing song (§7.2)
song view	SCALE	opens the drone view (chapter 9)
drone view	SHIFT + KIT	creates a drone track (chapter 9)
kit clip view, audition pad held	SHIFT + KIT	turns the row into a drone row (chapter 9)

11.3 Displays and messages

Display	Where	Meaning
UGAP, "USB audio gap"	7-segment, OLED	The computer got an empty USB audio packet: a 1 ms gap in the recording. At most every 10 s. Please report it (§4.6).
C-ON, C-OFF, C-AL	7-segment	The shortcut set the CPU monitor to On, Off or Alerts (§3.4).
-3.5, OFF	7-segment	A volume in dB; OFF at the very bottom (§5.7).
a blinking number	7-segment	Song change: loops remaining. Without blinking: bars. With a dot: beats (§7.2).
"Bars remaining 3"	OLED	A countdown in the title line, over everything: song change, or in Song view clips or a section armed to launch; popups appear below it (§7.2).
"Open the group first"	OLED	You tried to delete on a closed group line in the song browser (§7.1).
small animation, top right	OLED	The Deluge is loading or saving (§7.3).
1.2.1-mastertune- / v18.3-l2d	OLED	Settings → Firmware version (§1.4).

11.4 CPU monitor abbreviations

Shown	Where	Meaning
CPU 39%	OLED line	Load of the audio routine. From v17 on the real load, without busy waiting.
14V	OLED line	Voices sounding now.
QL	OLED line, live view	Quality lowered: the Deluge is behind (direness above 0) and computes more simply. Shown only while it happens.
VC	OLED line	Voices cut. Blinks and stays for 2 s after the last cut.
peak	live view, page	The most expensive audio window of the last second.
cut	live view	Voices cut by the CPU protection. The number that counts.
gap	live view, page	Longest pause between two runs of the audio routine. From about 2.9 ms on a dropout threatens.
SD	live view, page	Loads from the card, mean and longest duration.
direness	page, live view	0–14. Above 0 the quality is lowered; from about 80 samples behind voices are cut.

"Live view" is the line of `tools/deluge_profiler.py live` (§8.6), "page" is `tools/cpu_monitor.html` (§8.5).

11.5 Files on the card

File or folder	What mastertune keeps there
the firmware .bin (root folder)	Only for installing; leave only one there (§1.3).
CommunityFeatures.XML (root folder)	The community settings, the master tune (masterTune, in tenths of a Hz), the CPU monitor mode.
SONGS/	Songs, with the drone and drone tracks (chapter 9).
SYNTHS/	Synth presets, for example the ping-pong presets (§6.4).
recordings (WAV)	At a tuning other than 440 Hz with an mtun chunk (§2.3).
BASELINE - BACKUP/	Backups written by DelugeBaseline (§8.4).

11.6 Troubleshooting

Find your problem in its group. Each line gives the cause and what to do.

Installing and settings

The Deluge doesn't install the firmware.	Leave only one .bin in the root folder and hold SHIFT while switching on (§1.3).
It crashes or behaves strangely.	Install deluge-1.2.1-mastertune-v18-124aeaa2.bin, v18 without L2, and report what happened (§11.8).
The master tune is back at the old value after switching off.	It is saved when you leave the Settings menu. Leave it with BACK first (§2.1).
Output limiter, crossing guard and OLED brightness are back at their defaults.	You switched to v17 or 1.2.1 and back; they lose settings they don't know. Set them again (§7.6).

Tuning

An external synth doesn't follow the master tune.	It must evaluate RPN 1 (Channel Fine Tuning). A device switched on later gets the tuning after the next play start (§2.4).
A recording made at 432 Hz sounds off after editing it on the computer.	Some editors remove the mtun chunk; the copy then counts as 440 Hz material. Use the original recording (§2.3).

Load

QL or VC appear.	The Deluge can't keep up. See what costs CPU in §3.5: at 432 Hz convert the samples with DelugeTuner; use fewer living drone tones.
The CPU monitor shows 85–90% while nothing plays.	Up to v16 that was busy waiting, not overload. From v17 on the monitor shows the real load (§3.4).
The first hit of an unused kit row comes late.	The Kit RAM saver released its sample start. Switch it off if this bothers you (§3.3).

USB

The computer doesn't show the audio input.	Switch USB audio on, restart, and connect the Deluge as a USB device, not as a host (§4.2).
USB audio is off again after a restart.	Up to v15 the setting was saved only when you left the menu with BACK . From v16 on it is saved at once.
The Deluge isn't recognised or hangs with USB audio.	Unplug the USB cable, start the Deluge, switch USB audio off (§4.6).
The DAW lost the Deluge's MIDI ports.	With USB audio on, the computer sees a new device. Assign the ports again (§4.6).
UGAP appears.	A 1 ms gap under very high load. Please report it (§4.6).
deluge-editor says "needs community 1.3.0 or later".	The Deluge reports 1.2.1. Ignore it: opening and saving work (§8.7).
A file with umlauts in its name doesn't open over USB.	Use only A–Z, 0–9 and _ in names (§8.7).

Sound

A song with routing Parallel is quieter.	Since v18 the parallel sum has half the volume, −6 dB. Raise the song's volume (§5.2).
The filter whistles, or a tone follows the cutoff knob.	The resonance is so high that the filter oscillates by itself. Turn the resonance down (§5.3).
An old song's reverb sounds stiller.	Songs without the new settings load with Modulation 0. Set Modulation to 50 for the old sound (§5.10).
Echoes bend in pitch when the delay time changes.	Time change is on Tape; old songs with a modulated or automated delay rate load that way. Choose Fade (§5.11).
Hard sidechain pumping sounds softer.	Since v18 ducking takes at least 5 ms (§5.7).
The output clips.	Only the sum at the output clips, at 0 dBFS. Lower the song volume until the peak lies at −6 dBFS, or switch the Output limiter on (§5.8).
A click when switching the Output limiter.	The output moves once by 0.7 ms. This is expected (§5.8).

Tools

Windows warns when you start a tool.	The programs are not signed. Click More info, then Run anyway (§8.1).
DelugeRec names the take "... 1.2.1 v18.WAV" with v18.3.	DelugeRec v6 cuts the version at the dot. Use v7 or later (§8.2).
The profiler shows wrong function names.	Load the <code>.symbols.json</code> that lies next to the <code>.bin</code> you installed (§8.6).

11.7 Version history

Version	What it added
v2	Master tune 415.3–466.2 Hz; everything follows, recordings keep their tuning.
v3	Cheaper sample interpolation; sidechain without click.
v4	Gentler culling as in the community firmware; silent kits and audio tracks skip their effects.
v5	The arpeggiator from 1.3, with latch and ratchet bounce.
v6	A fixed number of ratchet notes, bounce without getting quieter.
v7	The SD card over USB for DEX and deluge-editor.
v8	USB audio: the Deluge's output as an audio input on the computer.
v9	Smarter sample streaming, Kit RAM saver.
v10	Reverb: new model Digital, Mutable and Freeverb repaired, HPF and LPF.
v11	Delay: clean repeats, no clicks on time changes, LPF and HPF in the feedback; no crackle when saving.
v12	Frequency drone with up to 16 tones.
v13	More performance, ping-pong roll, flicker-free dimming, faster saving, more precise MIDI, CPU monitor in words, drone polish.
v14	Reverb without wobble, delay without pitch jumps, countdown on song change, two card fixes.
v15	Living drone (Life, FM, Pulse), CPU monitor in one line, section launch by CC, USB MIDI without packet loss, clock outputs under external clock.
v16	Drone tracks, profiler, USB audio stays on after a restart.
v17	Much less load at moderate load (scheduling fix, minimum window, silent tracks), song browser with versions, CPU monitor shortcut, drone view without hangs, quieter HPF whistle; main file with L2 cache.
v18	Filters without rustle and with clean transitions, volume in 0.5 dB steps shown in dB, real shelving EQ, Output limiter, Filter crossing guard, OLED brightness, readable version, song name for DelugeRec.
v18.2	Filters, EQ and volume right after a silence. L2 variant only.
v18.3	Drive oversampled like 1.2.1, unknown settings and the reverb sidechain's sync level kept. L2 variant only.

11.8 Reporting bugs

mastertune is a personal fork. **Don't report its bugs to the community developers.** Report them at github.com/Giansn/deluge/issues, with:

- the version from Settings → Firmware version;
- what you did, exactly;
- what the display showed: frozen, a restart, or a message like "E...";
- for load problems, if you can, a CSV from the CPU monitor page (§8.5) or a `.jsonl` recording from the profiler (§8.6).